

E1084976250_1

The equation $2x^4 + x + y^3 + 4y + 1 = 0$ has no integer solutions

Size $h = 51$, length $l = 12$

Theorem 1. *The equation*

$$2x^4 + x + y^3 + 4y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 42 auxiliary polynomials $H_0, \dots, H_{41}$, 7 of which are constants, and 2 graphs on these vertices with 83 and 69 edges, respectively; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned}\beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2.\end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = 2x^4 + x + y^3 + 4y + 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= -x - y, \\ H_2 &= -y - 1, \\ H_3 &= -y + 1, \\ H_4 &= x - y, \\ H_5 &= -2x - y, \\ H_6 &= -x - y - 1, \\ H_7 &= -x - y + 1, \\ H_8 &= -y - 2, \\ H_9 &= x - y - 1, \\ H_{10} &= x - y + 1, \\ H_{11} &= 2x - y - 1, \\ H_{12} &= 2x - y + 1, \\ H_{13} &= -2x - y - 2, \\ H_{14} &= x^2 - y, \\ H_{15} &= x^2 + y - 1, \\ H_{16} &= x^2 + x + y, \\ H_{17} &= x^2 - x - y, \\ H_{18} &= x^2 + x - y, \\ H_{19} &= x^2 - x + y - 1, \\ H_{20} &= x^2 - x + y + 1, \\ H_{21} &= x^2 + x + y + 1, \\ H_{22} &= x^2 - 2x - y, \\ H_{23} &= x^2 - y - 2, \\ H_{24} &= x^2 + x - y - 1, \\ H_{25} &= x^2 - 2x + y + 1, \\ H_{26} &= x^2 + x + y - 2, \\ H_{27} &= x^2 + 2x + y - 1, \\ H_{28} &= x^2 + 2x + y + 1, \\ H_{29} &= x^2 - 2x - y - 1, \\ H_{30} &= 1, \\ H_{31} &= -1, \\ H_{32} &= 7, \\ H_{33} &= 2, \\ H_{34} &= 3, \\ H_{35} &= -x - y - 2, \\ H_{36} &= x^2 - y + 1, \\ H_{37} &= x^2 - x - y - 1, \end{aligned}$$

$$\begin{aligned}
H_{38} &= x^2 - 2x + y - 1, \\
H_{39} &= x^2 - x + y - 2, \\
H_{40} &= 21, \\
H_{41} &= -2.
\end{aligned}$$

There are 2 graphs on these vertices. The edges of the first graph are $\{0, 1\}, \{0, 9\}, \{0, 20\}, \{1, 7\}, \{1, 11\}, \{1, 22\}, \{1, 28\}, \{2, 5\}, \{2, 12\}, \{2, 14\}, \{2, 16\}, \{2, 23\}, \{2, 25\}, \{3, 4\}, \{3, 5\}, \{3, 9\}, \{3, 21\}, \{4, 5\}, \{4, 7\}, \{4, 11\}, \{4, 16\}, \{4, 19\}, \{4, 20\}, \{4, 24\}, \{5, 17\}, \{5, 18\}, \{5, 21\}, \{5, 27\}, \{5, 28\}, \{6, 10\}, \{6, 14\}, \{6, 20\}, \{6, 29\}, \{7, 9\}, \{7, 10\}, \{7, 13\}, \{7, 16\}, \{7, 21\}, \{7, 26\}, \{8, 29\}, \{9, 14\}, \{9, 16\}, \{9, 19\}, \{9, 21\}, \{9, 26\}, \{10, 15\}, \{11, 18\}, \{11, 24\}, \{11, 28\}, \{12, 16\}, \{12, 20\}, \{12, 21\}, \{13, 17\}, \{13, 19\}, \{14, 22\}, \{14, 29\}, \{15, 27\}, \{15, 28\}, \{16, 27\}, \{16, 29\}, \{17, 28\}, \{18, 24\}, \{19, 25\}, \{19, 27\}, \{20, 25\}, \{22, 23\}, \{23, 24\}, \{24, 29\}, \{26, 27\}, \{26, 28\}, \{27, 28\}, \{3, 31\}, \{4, 32\}, \{5, 33\}, \{13, 33\}, \{14, 33\}, \{19, 34\}, \{21, 31\}, \{22, 31\}, \{23, 33\}, \{24, 34\}, \{27, 34\}, \{29, 34\}$. The edges of the second graph are $\{0, 2\}, \{0, 7\}, \{0, 12\}, \{0, 13\}, \{0, 36\}, \{2, 3\}, \{2, 8\}, \{2, 9\}, \{2, 14\}, \{2, 37\}, \{2, 24\}, \{2, 25\}, \{3, 9\}, \{3, 11\}, \{3, 12\}, \{3, 16\}, \{3, 28\}, \{4, 7\}, \{4, 11\}, \{4, 14\}, \{4, 16\}, \{4, 20\}, \{6, 7\}, \{6, 35\}, \{6, 13\}, \{6, 14\}, \{6, 20\}, \{6, 29\}, \{7, 13\}, \{7, 16\}, \{7, 17\}, \{7, 26\}, \{9, 12\}, \{9, 14\}, \{9, 16\}, \{9, 17\}, \{9, 19\}, \{9, 24\}, \{9, 26\}, \{11, 12\}, \{12, 36\}, \{12, 28\}, \{13, 17\}, \{13, 19\}, \{14, 24\}, \{14, 29\}, \{16, 29\}, \{17, 28\}, \{36, 27\}, \{19, 20\}, \{19, 24\}, \{19, 38\}, \{19, 25\}, \{19, 39\}, \{20, 37\}, \{37, 24\}, \{37, 29\}, \{38, 27\}, \{25, 28\}, \{26, 27\}, \{26, 28\}, \{0, 33\}, \{3, 34\}, \{4, 40\}, \{7, 34\}, \{9, 41\}, \{36, 33\}, \{27, 33\}, \{29, 31\}$. For a vertex i , we write $j \sim_g i$ if $\{i, j\}$ is an edge of the g -th graph. For every non-constant vertex i and every graph g in which i has a neighbour, Table 1 gives a positive integer $m_{g,i}$ and polynomials $W_{g,i}$, $A_{g,i}$ and $B_{g,i}$ with integer coefficients such that

$$m_{g,i}^2 \prod_{j \sim_g i} H_j - W_{g,i}^2 = A_{g,i}F + B_{g,i}H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij}F + \beta_{ij}H_i + \gamma_{ij}H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -108x^8 - 108x^5 - 108x^4 - 27x^2 - 54x - 283.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 35, 36, 37, 38, 39\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(7 + t)$ and the $\varphi_i(7 + t)$ as polynomials in t gives

in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-6-t)$ and the $\varphi_i(-6-t)$. Hence, for $x \geq 7$ and for $x \leq -6$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-5 \leq x \leq 6$. Checking the residues of y , we find that for $x = -1$ and $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x = -3$ and $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x = 1$ and $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , for $x = 2$ and $x = 6$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , for $x = -5$ and $x = 5$, we have $F(x, y) \not\equiv 0 \pmod{19}$ for every y , for $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , and for $x = -4$, we have $F(x, y) \not\equiv 0 \pmod{37}$ for every y . Hence (1) has no solution with $-5 \leq x \leq 6$.

For a place v , put $B_v^{(g)} = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges of the g -th graph. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{30, 32, 33, 34, 40\}$ and $H_i < 0$ for $i \in \{31, 41\}$. The other signs are given in Table 3 for $x \geq 7$ and for $x \leq -6$. In both cases, counting the edges of each graph whose two values are negative gives $(B_\infty^{(g)})_g = (1, 1)$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11\}$ be the set of primes dividing 2, the $m_{g,i}$, the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p^{(g)} = 1$ for every g . For an edge $\{i, j\}$ of the g -th graph such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$ of the g -th graph, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim_g i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim_g i} H_j$ is not divisible by p . By (2), $m_{g,i}^2 \Pi_i \equiv W_{g,i}^2 \pmod{p}$, and $p \nmid m_{g,i}$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p^{(g)} = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p^{(g)} = (-1)^{N_g}$, where $N_g = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges of the g -th graph. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges of every graph, then $N_g = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so $B_p^{(g)}$ is determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of the vector $(B_p^{(g)})_g$; every integer solution lies in one of them. The result is $(B_2^{(g)})_g \in \{(1, -1), (-1, 1), (-1, -1)\}$, $(B_3^{(g)})_g \in \{(1, 1)\}$, $(B_5^{(g)})_g \in \{(1, 1)\}$, $(B_7^{(g)})_g \in \{(1, 1)\}$, $(B_{11}^{(g)})_g \in \{(1, 1)\}$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v^{(g)} = 1$ for every g . By Step 3, the factors with $p \notin S$ are 1, so the componentwise product of the vectors $(B_p^{(g)})_g$ with $p \in S$ is $(B_\infty^{(g)})_g = (1, 1)$ by Step 2. But by Step 4, this product is $(1, -1)$, $(-1, 1)$, or $(-1, -1)$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$g = 1, i = 0, m = 1, \text{neighbours } 1, 9, 20$	
W	$2x^3 - x^2 + 1$
A	$-2x^2 + 2x - 1$
B	$-2x^2y^2 - x^2y - 8x^2 + 2xy^2 + xy + 8x - 2y^2 - 2y - 5$
$g = 1, i = 1, m = 2, \text{neighbours } 0, 7, 11, 22, 28$	
W	$-2x^3$
A	$4x^2y - 2x^2 + 2xy^2 - 6xy - 2y^3$
B	$20x^3y + 24x^2y^2 - 16x^2y - 2x^2 + 4xy^4 - 14xy^3 + 2xy^2 - 20xy - 2x - 2y^5$ $-4y^4 - 12y^3 + 2y^2 + 4y$
$g = 1, i = 2, m = 2, \text{neighbours } 5, 12, 14, 16, 23, 25$	
W	$-2x^3 + 6x^2 - 8$
A	$-8x^6 + 4x^5 + 2x^4y^2 - 2x^4y + 28x^4 - 2x^3y^2 - 6x^3y - 8x^3$ $+4x^2y^3 + 10x^2y^2 + 10x^2y - 28x^2 + 3xy^2 - xy + 4x - y^5 - 3y^4 - 12y^3$ $+10y^2 - 16y - 20$
B	$-2x^3y^4 - 8x^3y^3 - 8x^3y^2 - 32x^3y - 4x^3 + 4x^2y^5 + 6x^2y^4$ $+40x^2y^3 + 28x^2y^2 + 33x^2y - 120x^2 + 6xy^4 - 2xy^3 + 18xy^2 + 15xy - 16x$ $-y^7 - 2y^6 - 18y^5 + 7y^4 - 70y^3 + 86y^2 - 140y + 44$
$g = 1, i = 3, m = 2, \text{neighbours } 31, 4, 5, 9, 21$	
W	$2x^2 - 8$
A	$4x - 6y - 2$
B	$-4x^2y^2 + 4x^2y + 20x^2 - 8xy - 2x - 10y^3 - 20y^2 - 48y - 62$
$g = 1, i = 4, m = 8, \text{neighbours } 32, 3, 5, 7, 11, 16, 19, 20, 24$	
W	$150x^3 + 108x^2 + 114x + 6$
A	$-896x^7y + 896x^7 - 896x^6y^2 + 2240x^6y - 1344x^6 + 224x^5y^3$ $-1344x^5y^2 + 3360x^5y - 2240x^5 + 224x^4y^4 - 2016x^4y^3 + 5152x^4y^2$ $-7840x^4y + 4480x^4 + 896x^3y^4 - 2016x^3y^3 + 6720x^3y^2 - 4032x^3y$ $-1568x^3 + 896x^2y^5 - 1792x^2y^4 + 4592x^2y^3 - 12544x^2y^2$ $+10640x^2y - 13042x^2 - 112xy^6 + 1120xy^5 - 5376xy^4 + 11872xy^3 - 8624xy^2$ $-2240xy - 12840x - 112y^7 + 224y^6 - 224y^5 - 1624y^4 - 7924y^3 + 9618y^2 + 11652y$ -36
B	$1568x^3y^6 - 2464x^3y^5 + 17136x^3y^4 - 21784x^3y^3 + 12796x^3y^2$ $-26664x^3y - 49600x^3 + 672x^2y^7 + 448x^2y^6 - 784x^2y^5 - 2520x^2y^4$ $-10612x^2y^3 - 12776x^2y^2 - 44112x^2y - 14502x^2 - 224xy^8 + 1344xy^7$ $-6608xy^6 + 17416xy^5 - 39620xy^4 + 37562xy^3 - 61472xy^2 + 24338xy + 12486x$ $-112y^9 + 224y^8 - 224y^7 - 840y^6 - 9940y^5 + 2898y^4 - 20324y^3 + 30512y^2$ $+55778y + 11508$
$g = 1, i = 5, m = 2, \text{neighbours } 33, 2, 3, 4, 17, 18, 21, 27, 28$	
W	$-1770x^3 - 348x^2 - 1443x + 213$
A	$4x^7y^2 - 4x^7 - 4x^6y^3 + 20x^6y^2 + 4x^6y - 20x^6 - 16x^5y^3$ $+32x^5y^2 + 16x^5y - 32x^5 - 4x^4y^4 - 32x^4y^3 + 14x^4y^2 + 32x^4y$ $-10x^4 - 2x^3y^5 - 8x^3y^4 - 60x^3y^3 - 28x^3y^2 + 62x^3y + 36x^3$ $+2x^2y^6 - 2x^2y^5 + 10x^2y^4 - 92x^2y^3 - 74x^2y^2 + 94x^2y$ $-1566388x^2 + 8xy^6 + 16xy^5 + 62xy^4 - 88xy^3 - 107xy^2 + 72xy - 615923x - 4y^7$ $+84y^6 - 680y^5 + 6074y^4 - 51125y^3 + 428119y^2 - 456529y - 45369$
B	$-6x^3y^7 + 60x^3y^6 - 734x^3y^5 + 5980x^3y^4 - 51104x^3y^3$ $+428220x^3y^2 - 456494x^3y + 2569310x^3 + 2x^2y^8 - 34x^2y^7$ $+330x^2y^6 - 3070x^2y^5 + 25444x^2y^4 - 214184x^2y^3 + 228340x^2y^2$ $-1284497x^2y - 658018x^2 + 16xy^8 - 152xy^7 + 1498xy^6 - 12736xy^5$ $+106999xy^4 - 897598xy^3 + 642330xy^2 - 2803688xy - 124155x - 4y^9 + 84y^8$ $-688y^7 + 6414y^6 - 53777y^5 + 451719y^4 - 654947y^3 + 1615990y^2 - 1397997y$ -638005
$g = 1, i = 6, m = 2, \text{neighbours } 10, 14, 20, 29$	
W	$-2x^3 + 2x^2$
A	$2x^3 - 2x^2y - 6x^2 + 4xy + 2x + 4y^2 + 6y + 2$
B	$4x^3y^2 + 16x^3y + 6x^3 - 2x^2y^3 - 8x^2y^2 - 12x^2y - 2x^2 - 4xy^2 + 2x$ $+4y^4 + 6y^3 + 16y^2 + 8y + 2$
$g = 1, i = 7, m = 2, \text{neighbours } 1, 4, 9, 10, 13, 16, 21, 26$	
W	$10x^3 - 18x^2 + 20x + 12$
A	$4x^7 - 6x^6y + 16x^6 - 4x^5y^2 - 14x^5y + 16x^5 + 8x^4y^3 - 30x^4y^2$ $-2x^4y - 14x^4 + 18x^3y^3 - 52x^3y^2 - 5x^3y - 46x^3 - 2x^2y^5$

	$+23x^2y^4 - 8x^2y^3 - 2x^2y^2 - 40x^2y - 90x^2 - 4xy^5 + 55xy^4 - 20xy^3$
	$+65xy^2 - 30xy + 187x - 9y^6 + 31y^5 - 9y^4 + 50y^3 + 28y^2 + 102y - 217$
B	$2x^3y^6 + 16x^3y^5 - 60x^3y^4 - 54x^3y^2 + 105x^3y + 190x^3$
	$-2x^2y^7 + 28x^2y^6 - 22x^2y^5 - 2x^2y^4 - 174x^2y^3 - 225x^2y^2$
	$-354x^2y - 442x^2 + 5xy^7 + 46xy^6 + 16xy^5 + 170xy^4 - 19xy^3 + 56xy^2 - 358xy$
	$-377x - 9y^8 + 26y^7 - 15y^6 + 130y^5 + 133y^4 + 442y^3 + 403y^2 + 839y + 73$
$g = 1, i = 8, m = 1, \text{ neighbours } 29$	
W	$x - 1$
A	0
B	1
$g = 1, i = 9, m = 16, \text{ neighbours } 0, 3, 7, 14, 16, 19, 21, 26$	
W	$-58x^3 - 70x^2 + 320x - 136$
A	$-128x^7y^2 + 128x^7y - 128x^6y^3 + 128x^6y - 640x^5y^3 + 1152x^5y^2$
	$-512x^5y - 384x^4y^4 + 512x^4y^3 + 448x^4y^2 - 576x^4y + 64x^3y^5$
	$-832x^3y^4 + 2880x^3y^3 - 2240x^3y^2 + 128x^3y + 64x^2y^6$
	$-256x^2y^5 + 1216x^2y^4 + 128x^2y^3 - 2112x^2y^2 + 960x^2y - 1682x^2$
	$+320xy^6 - 320xy^5 + 1984xy^4 - 3520xy^3 + 608xy^2 + 928xy - 4060x + 128y^7$
	$+416y^6 - 656y^5 - 1720y^4 - 3700y^3 + 5362y^2 + 8620y + 4958$
B	$128x^3y^7 - 832x^3y^6 + 2336x^3y^5 + 1136x^3y^4 + 2472x^3y^3$
	$-4580x^3y^2 - 17560x^3y + 22304x^3 + 64x^2y^8 + 128x^2y^7 + 96x^2y^6$
	$+3856x^2y^5 - 4072x^2y^4 + 3332x^2y^3 - 18044x^2y^2 + 3144x^2y$
	$+53010x^2 + 448xy^8 - 480xy^7 + 2416xy^6 - 1880xy^5 - 2596xy^4 - 3494xy^3$
	$-16980xy^2 + 60994xy - 62688x + 128y^9 + 288y^8 - 688y^7 + 1784y^6 - 8716y^5$
	$+6030y^4 - 12650y^3 + 34844y^2 + 4998y + 23454$
$g = 1, i = 10, m = 1, \text{ neighbours } 6, 7, 15$	
W	$2x^2 + 2x$
A	0
B	$-3x^3 - x^2y - 5x^2 - 3xy - x - y^2 + 1$
$g = 1, i = 11, m = 2, \text{ neighbours } 1, 4, 18, 24, 28$	
W	$10x^3 - 12x^2 + 15x - 4$
A	$-2x^4 - 8x^3 + 2x^2y^2 + 2x^2y - 60x^2 + 8xy^2 + 8xy + 119x - 5y^3 + 8y^2 - 36y - 45$
B	$4x^3y^3 + 4x^3y^2 + 50x^3y - 168x^3 + 2x^2y^4 + 28x^2y^2 - 42x^2y$
	$+172x^2 - 2xy^4 + 28xy^3 - 24xy^2 + 178xy - 104x - 5y^5 + 9y^4 - 73y^3 + 51y^2$
	$-187y - 29$
$g = 1, i = 12, m = 2, \text{ neighbours } 2, 16, 20, 21$	
W	$2x^2 + 6x + 4$
A	$-2x^2y - 2x^2 - 2xy - 2x - 2y^2 - 4y - 2$
B	$-4x^3y^2 - 4x^3y - 2x^3 - 2x^2y^3 - 4x^2y^2 - 4x^2y - 12x^2 - 6xy^3$
	$-8xy^2 - 6xy - 20x - 2y^4 - 2y^3 - 6y - 14$
$g = 1, i = 13, m = 1, \text{ neighbours } 33, 7, 17, 19$	
W	$-2x^2 - 7x - 3$
A	$-x - y + 1$
B	$-2x^2y + 16x^2 + xy^2 - 6xy + 15x - y^3 + y^2 - 2y + 5$
$g = 1, i = 14, m = 1, \text{ neighbours } 33, 2, 6, 9, 22, 29$	
W	$-4x^2 - 2x$
A	$4x^3 + x^2y + x^2 - 4xy - 4x - y^2 - 9y$
B	$-8x^5 - 8x^3y - 2x^2y^3 - 8x^2y^2 + 14x^2y - 16x^2 + 4xy^3 + 24xy^2 + 19xy$
	$-9x + y^4 - y^3 + 8y^2 - 29y - 7$
$g = 1, i = 15, m = 1, \text{ neighbours } 10, 27, 28$	
W	$-2x^2 - 2x$
A	0
B	$x^3 - x^2y + x^2 - 3xy + x - y^2 + 1$
$g = 1, i = 16, m = 16, \text{ neighbours } 2, 4, 7, 9, 12, 27, 29$	
W	$-16x^5 + 160x^3 + 96x^2 - 64x + 16$
A	$-128x^6 + 256x^4y + 2816x^4 + 384x^3y^3 - 352x^3y^2 - 1784x^3y + 4866x^3$
	$-64x^2y^3 + 512x^2y^2 - 640x^2y - 15296x^2 + 384xy^4 - 288xy^3 + 744xy^2$
	$+4218xy - 15170x - 448y^4 + 736y^3 - 8y^2 - 16962y - 256$
B	$-768x^5y^3 - 64x^5y^2 + 2032x^5y - 7300x^5 + 768x^4y^3 + 64x^4y^2$
	$-2032x^4y + 7300x^4 + 768x^3y^4 - 128x^3y^3 - 2048x^3y^2 + 7808x^3y$
	$-5120x^3 - 256x^2y^5 - 896x^2y^4 + 2912x^2y^2 + 4728x^2y + 9726x^2$

$$\begin{aligned}
& -384xy^6 - 160xy^5 - 1544xy^4 - 4290xy^3 - 2720xy^2 - 18184xy + 5568x + 704y^6 \\
& -736y^5 + 1032y^4 + 14466y^3 + 320y^2 + 67856y + 17730 \\
g = 1, i = 17, m = 8, \text{ neighbours } 5, 13, 28 \\
W & 8x^5 - 16x^4 + 8x^3 + 8x^2 + 24x \\
A & -32x^6 + 128x^5 - 192x^4 + 48x^3y^2 - 68x^3y + 307x^3 + 16x^2y^3 + 64x^2y \\
& -144x^2 - 64xy^3 - 12xy^2 - 239xy + 189x - 16y^4 - 16y^3 - 28y^2 + 61y \\
B & -96x^5y^2 + 136x^5y - 454x^5 - 96x^4y^2 + 136x^4y - 454x^4 - 96x^3y^3 \\
& + 64x^3y^2 - 352x^3y - 128x^3 + 32x^2y^4 + 32x^2y^3 + 208x^2y^2 \\
& - 92x^2y - 435x^2 - 48xy^5 + 4xy^4 - 419xy^3 + 80xy^2 - 1188xy - 214x - 16y^6 \\
& - 16y^5 - 92y^4 - 19y^3 - 192y^2 + 24y - 67 \\
g = 1, i = 18, m = 40, \text{ neighbours } 5, 11, 24 \\
W & -8x^5 + 32x^2 - 48x + 24 \\
A & -32x^6 + 16x^3y^2 - 100x^3y + 281x^3 + 16x^2y^3 + 64x^2y - 368x^2 - 12xy^2 \\
& + 75xy + 215x - 16y^4 + 48y^3 - 132y^2 + 169y - 576 \\
B & -32x^5y^2 + 200x^5y - 18x^5 + 32x^4y^2 - 200x^4y + 18x^4 - 32x^3y^3 \\
& + 192x^3y^2 + 32x^3y - 64x^3 + 32x^2y^4 - 32x^2y^3 - 144x^2y^2 \\
& - 252x^2y - 6489x^2 - 16xy^5 + 36xy^4 - 137xy^3 + 560xy^2 - 900xy + 6448x - 16y^6 \\
& + 48y^5 - 196y^4 + 345y^3 + 544y^2 + 3744y - 535 \\
g = 1, i = 19, m = 2, \text{ neighbours } 34, 4, 9, 13, 25, 27 \\
W & -12x^3 + 18x^2 + 12x \\
A & 12x^3 + 18x^2y - 72x^2 - 6xy + 12x + 18y^2 - 114y + 96 \\
B & -48x^5 - 48x^4 + 48x^3y + 432x^3 - 12x^2y^3 + 156x^2y + 48x^2 - 24xy^3 - 246xy \\
& + 36x - 30y^4 + 48y^3 - 132y^2 + 342y + 96 \\
g = 1, i = 20, m = 8, \text{ neighbours } 0, 4, 6, 12, 25 \\
W & 24x^5 - 32x^4 + 40x^3 - 104x^2 + 80x - 64 \\
A & -288x^6 + 768x^5 - 1472x^4 - 240x^3y^2 - 36x^3y + 2121x^3 + 144x^2y^3 \\
& + 576x^2y - 6288x^2 - 384xy^3 + 60xy^2 - 1463xy + 5776x + 144y^4 - 384y^3 + 812y^2 \\
& - 4843y - 5159 \\
B & 480x^5y^2 + 72x^5y + 3598x^5 + 480x^4y^2 + 72x^4y + 3598x^4 - 480x^3y^3 \\
& - 192x^3y^2 - 3616x^3y + 5664x^3 - 288x^2y^4 + 1280x^2y^3 - 2192x^2y^2 \\
& + 8260x^2y - 9577x^2 + 240xy^5 + 228xy^4 + 519xy^3 + 2064xy^2 - 3004xy + 10686x \\
& - 144y^6 + 528y^5 - 1852y^4 + 8151y^3 - 5920y^2 + 24416y + 1063 \\
g = 1, i = 21, m = 2, \text{ neighbours } 31, 3, 5, 7, 9, 12 \\
W & 6x^3 + 8x^2 + 4x + 8 \\
A & -18x^2 - 14x + 4y^2 - 14y - 2 \\
B & -68x^3 - 8x^2y^2 + 44x^2y - 56x^2 + 8xy^2 + 6x + 6y^3 - 20y^2 + 80y - 62 \\
g = 1, i = 22, m = 1, \text{ neighbours } 31, 1, 14, 23 \\
W & 2x^2 - 2x \\
A & 0 \\
B & x^3 + x^2y - 2x^2 + xy + 2x - y^2 - 2y \\
g = 1, i = 23, m = 1, \text{ neighbours } 33, 2, 22, 24 \\
W & 2x^2 - 2 \\
A & 0 \\
B & -2x^2y - 6x^2 + 2xy + 2x + 2y^2 + 2 \\
g = 1, i = 24, m = 2, \text{ neighbours } 34, 4, 11, 18, 23, 29 \\
W & -6x^3 + 6x \\
A & 12x^4 - 42x^3y - 48x^3 + 6x^2y^2 - 12x^2y - 72x^2 + 48xy^2 + 144xy + 60x - 6y^3 \\
& - 48y^2 - 42y \\
B & 48x^5y + 60x^5 - 48x^4y - 60x^4 + 48x^3y^2 + 120x^3y + 72x^3 - 36x^2y^3 \\
& - 96x^2y^2 - 78x^2y + 108x^2 + 42xy^4 + 48xy^3 + 114xy^2 + 192xy + 60x - 6y^5 \\
& - 30y^4 + 12y^3 - 150y^2 - 42y \\
g = 1, i = 25, m = 1, \text{ neighbours } 2, 19, 20 \\
W & -x^2 + 2x \\
A & 0 \\
B & -x^2y - 2x^2 + 2x - y^2 + 1 \\
g = 1, i = 26, m = 1, \text{ neighbours } 7, 9, 27, 28 \\
W & -x^3 - 3x^2 + x + 3 \\
A & -x^2 - x - y + 2 \\
B & -6x^3 + x^2y^2 - 2x^2 + 3xy^2 + 6xy + 7x + 2y^3 + 2y^2 + 6y + 5 \\
g = 1, i = 27, m = 64, \text{ neighbours } 34, 5, 15, 16, 19, 26, 28
\end{aligned}$$

W	$384x^3 - 384x$
A	$-12288x^7 - 6144x^6y - 36864x^6 - 79872x^5y + 24576x^5 - 30720x^4y^2$ $-135168x^4y + 165888x^4 + 33792x^3y^3 - 203136x^3y^2 + 475824x^3y$ $+759354x^3 + 3072x^2y^4 - 43008x^2y^3 - 153600x^2y^2 + 595968x^2y$ $-251904x^2 + 39936xy^4 - 304512xy^3 + 594480xy^2 - 470838xy - 3526434x + 3072y^5$ $-89088y^4 + 123648y^3 - 381024y^2 - 1101588y + 1444980$
B	$-55296x^5y^3 + 13056x^5y^2 - 515424x^5y - 1322100x^5 + 110592x^4y^3$ $-26112x^4y^2 + 1030848x^4y + 2644200x^4 + 55296x^3y^4 - 172032x^3y^3$ $+552960x^3y^2 - 159744x^3y - 92160x^3 + 24576x^2y^5 - 30720x^2y^4$ $+46080x^2y^3 + 295296x^2y^2 - 905904x^2y - 618042x^2 - 33792xy^6$ $+98688xy^5 - 595632xy^4 - 78906xy^3 - 1694784xy^2 - 2696760xy + 808506x - 3072y^7$ $+86016y^6 - 62208y^5 + 708960y^4 + 1392756y^3 + 1311360y^2 + 6123312y + 1444980$
$g = 1, i = 28, m = 128, \text{neighbours } 1, 5, 11, 15, 17, 26, 27$	
W	$-256x^5 - 1024x^4 - 1280x^3 - 1536x^2$
A	$32768x^7 + 32768x^6y + 16384x^6 - 8192x^5y^2 + 106496x^5y - 458752x^5$ $-8192x^4y^3 + 40960x^4y^2 - 245760x^4y - 983040x^4 + 4096x^3y^4$ $-25088x^3y^3 + 115264x^3y^2 - 188936x^3y - 754119x^3 - 32768x^2y^4$ $+16384x^2y^3 - 495616x^2y^2 + 765952x^2y - 2285568x^2 + 4096xy^5$ $-61952xy^4 + 67648xy^3 - 59592xy^2 + 386129xy - 3029675x - 32768y^5 - 31744y^4$ $-405632y^3 - 426096y^2 - 1501282y - 1481842$
B	$-8192x^5y^4 - 48128x^5y^3 - 279680x^5y^2 - 588784x^5y - 1031282x^5$ $+16384x^4y^4 + 96256x^4y^3 + 559360x^4y^2 + 1177568x^4y + 2062564x^4$ $+8192x^3y^5 + 40960x^3y^4 + 237568x^3y^3 + 344064x^3y^2 + 516096x^3y$ $-49152x^3 + 8192x^2y^6 - 8192x^2y^5 + 274432x^2y^4 + 451072x^2y^3$ $+220608x^2y^2 + 1557000x^2y + 737735x^2 - 4096xy^7 + 512xy^6 - 33344xy^5$ $-722424xy^4 - 248121xy^3 - 1951136xy^2 - 207596xy + 1547833x + 32768y^7 + 15360y^6$ $+472192y^5 + 130032y^4 + 3074674y^3 + 484416y^2 + 5946808y + 1481842$
$g = 1, i = 29, m = 2, \text{neighbours } 34, 6, 8, 14, 16, 24$	
W	$12x^4 - 18x^3 + 6x$
A	$-72x^4 - 18x^3y + 249x^3 + 6x^2y^2 + 30x^2y - 126x^2 + 6xy^2 + 69xy - 129x$ $-6y^3 + 72y^2 + 66y - 12$
B	$48x^5y - 42x^5 + 96x^4y - 84x^4 + 48x^3y^2 + 108x^3y + 24x^3 + 72x^2y^3$ $-12x^2y^2 + 318x^2y + 3x^2 + 18xy^4 - 93xy^3 + 78xy^2 - 348xy - 117x - 6y^5$ $+66y^4 - 72y^3 + 282y^2 + 30y - 12$
$g = 2, i = 0, m = 1, \text{neighbours } 33, 2, 7, 12, 13, 36$	
W	$-4x^3 + 2x^2 + 2x$
A	$-8x^2 - 4xy + 4x - 4y^2 - 6y + 4$
B	$-2x^3y^2 + 2x^3 - 2x^2y^3 - 18x^2y^2 + 2x^2y - 26x^2 - 2xy^3 - 6xy^2$ $-14xy + 16x - 2y^4 - 6y^3 - 20y^2 - 24y + 16$
$g = 2, i = 2, m = 2, \text{neighbours } 0, 3, 8, 9, 14, 37, 24, 25$	
W	$2x^3 - 4x^2$
A	$-2x^5y^3 - 2x^5y^2 + 4x^5y + 2x^4y^4 + 8x^4y^3 + 2x^4y^2 - 12x^4y$ $-4x^2y^5 - 24x^2y^4 - 27x^2y^3 + 25x^2y^2 + 30x^2y - 2x^2 + xy^6$ $+13xy^5 + 39xy^4 + 19xy^3 - 44xy^2 - 28xy + 8x - y^7 - 4y^6 + 5y^5 + 13y^4 - 14y^3$ $-5y^2 + 6y - 8$
B	$32x^3y^5 + 84x^3y^4 - 16x^3y^3 - 107x^3y^2 + 8x^3y - 2x^3 - 4x^2y^7$ $-28x^2y^6 - 46x^2y^5 - 52x^2y^4 - 7x^2y^3 + 133x^2y^2 - 4x^2y + 6x^2$ $+xy^8 + 12xy^7 + 18xy^6 + 2xy^5 + 80xy^4 + 44xy^3 - 147xy^2 + 10xy - y^9 - 3y^8$ $+8y^7 + 8y^6 + 26y^5 + 34y^4 - 99y^3 + 29y^2 - 18y - 8$
$g = 2, i = 3, m = 2, \text{neighbours } 34, 2, 9, 11, 12, 16, 28$	
W	$8x^3 + 20x^2 + 8x$
A	$-24x^3y - 24x^3 + 48x^2y^2 - 80x^2 - 30xy^3 + 42xy^2 + 78xy - 154x + 6y^4$ $+18y^3 + 54y^2 + 114y - 192$
B	$60x^3y^3 + 120x^3y^2 + 36x^3y - 168x^3 + 24x^2y^4 + 24x^2y^3$ $+178x^2y^2 + 340x^2y + 146x^2 - 30xy^5 + 36xy^4 + 84xy^3 + 206xy^2 + 710xy$ $+334x + 6y^6 + 12y^5 + 54y^4 + 222y^3 + 288y^2 + 834y + 192$
$g = 2, i = 4, m = 2, \text{neighbours } 40, 7, 11, 14, 16, 20$	
W	$2x^3 - 22x^2 + 28x + 10$
A	$-84x^4 - 42x^3y + 126x^3 + 42x^2y^2 - 84x^2y - 44x^2 - 42xy^2 + 126xy + 2x$ $+84y^3 + 42y^2 - 62y - 100$
B	$84x^3y^2 + 166x^3y - 186x^3 + 42x^2y^4 + 42x^2y^3 + 40x^2y^2 - 144x^2y$

$$+1026x^2 + 42xy^4 + 168xy^3 + 528xy^2 + 824xy - 302x + 84y^5 + 126y^4 + 358y^3 + 68y^2 - 290y - 462$$

$g = 2, i = 6, m = 2$, neighbours 7, 35, 13, 14, 20, 29

W $-2x^3 + 2x^2$

A $-4x^5 - 10x^4y + 4x^4 - 8x^3y^2 + 20x^3y + 20x^3 - 2x^2y^3 + 28x^2y^2 + 22x^2y - 16x^2 + 16xy^3 - 9xy + 4y^4 - 4y^3 + 7y^2 + 35y + 20$

B $-6x^3y^4 + 4x^3y^3 - 58x^3y^2 - 8x^3y + 28x^3 - 2x^2y^5 + 14x^2y^4 - 4x^2y^3 + 86x^2y^2 + 98x^2y + 12xy^5 + 12xy^4 + 76xy^3 + 8xy^2 - 77xy + 4y^6 - 4y^5 + 47y^4 + 4y^3 + 36y^2 + 79y + 20$

$g = 2, i = 7, m = 2$, neighbours 34, 0, 4, 6, 13, 16, 17, 26

W $-10x^3 - 16x^2 + 30x - 36$

A $-12x^5y - 6x^4y^2 - 36x^4y + 12x^3y^3 - 12x^3y^2 + 6x^2y^4 + 24x^2y^3 + 36x^2y^2 + 78x^2y - 50x^2 + 24xy^4 + 12xy^3 + 99xy^2 + 60xy - 160x + 12y^5 + 6y^4 + 21y^3 - 39y^2 - 158y - 46$

B $6x^3y^5 + 12x^3y^4 + 6x^3y^3 - 228x^3y^2 - 280x^3y - 436x^3 + 6x^2y^6 + 30x^2y^5 + 168x^2y^4 + 222x^2y^3 + 196x^2y^2 + 282x^2y - 726x^2 + 12xy^6 + 54xy^5 + 174xy^4 + 132xy^3 + 409xy^2 + 946xy + 1116x + 12y^7 + 30y^6 + 111y^5 + 60y^4 - 56y^3 - 237y^2 - 908y - 1250$

$g = 2, i = 8, m = 1$, neighbours 2

W -1

A 0

B 1

$g = 2, i = 9, m = 32$, neighbours 41, 2, 3, 12, 14, 16, 17, 19, 24, 26

W $82x^3 - 466x^2 - 192x + 232$

A $-2048x^9y^2 + 2048x^9 + 1024x^8y^3 - 3072x^8y^2 - 1024x^8y + 3072x^8 + 1024x^7y^3 + 11264x^7y^2 - 1024x^7y - 11264x^7 - 4096x^6y^3 + 15360x^6y^2 + 4096x^6y - 15360x^6 + 1024x^5y^5 + 5120x^5y^4 - 4608x^5y^3 - 23040x^5y^2 + 3584x^5y + 17920x^5 - 512x^4y^6 - 1536x^4y^5 + 7680x^4y^4 + 512x^4y^3 - 31744x^4y^2 + 1024x^4y + 24576x^4 - 512x^3y^6 - 7680x^3y^5 - 7680x^3y^4 - 9216x^3y^3 + 9216x^3y^2 + 16896x^3y - 1024x^3 + 2048x^2y^6 - 4096x^2y^5 - 8704x^2y^4 + 768x^2y^3 + 19200x^2y^2 + 3328x^2y - 15906x^2 - 512xy^8 - 2560xy^7 - 3584xy^6 + 7680xy^5 + 9216xy^4 + 34048xy^3 + 13056xy^2 - 39168xy + 20036x + 256y^9 + 1152y^8 + 3264y^7 + 32y^6 + 15792y^5 + 26184y^4 + 6988y^3 - 21566y^2 - 87636y - 53682$

B $-768x^3y^8 - 6016x^3y^7 - 11840x^3y^6 - 21344x^3y^5 - 34960x^3y^4 + 62312x^3y^3 + 55420x^3y^2 + 87720x^3y - 97760x^3 - 256x^2y^9 + 896x^2y^8 - 6080x^2y^7 + 3168x^2y^6 - 2544x^2y^5 + 41176x^2y^4 + 65764x^2y^3 + 71716x^2y^2 - 24120x^2y - 297822x^2 - 256xy^{10} - 1408xy^9 - 576xy^8 + 160xy^7 + 21872xy^6 + 48616xy^5 + 97980xy^4 + 47914xy^3 - 20xy^2 - 226574xy - 122592x + 256y^{11} + 896y^{10} + 3392y^9 + 3552y^8 + 20304y^7 + 9272y^6 + 73204y^5 + 19614y^4 - 59258y^3 - 69604y^2 - 302506y + 142$

$g = 2, i = 11, m = 1$, neighbours 3, 4, 12

W $2x - 2$

A 0

B $-xy - x + y^2 - 3y + 4$

$g = 2, i = 12, m = 2$, neighbours 0, 3, 9, 11, 36, 28

W $-4x^3 - 20x^2 - 52x - 24$

A $4x^2y^2 - 4x^2y - 8x^2 - 6xy^3 + 8xy^2 - 2xy - 80x + 2y^4 - 2y^3 - 2y^2 - 46y - 48$

B $-8x^3y^3 + 8x^3y^2 + 48x^3y - 360x^3 - 8x^2y^3 + 30x^2y^2 - 202x^2y - 952x^2 - 2xy^5 + 10xy^4 + 14xy^3 - 126xy^2 - 352xy - 1312x + 2y^6 + 4y^5 + 14y^4 - 46y^3 - 112y^2 - 294y - 528$

$g = 2, i = 13, m = 2$, neighbours 0, 6, 7, 17, 19

W $-4x^3 - 18x^2 - 20x - 6$

A $-2x^2y - 8x^2 - 4xy^2 + 4xy - 72x - 2y^3 + 4y^2 - 22y - 64$

B $-4x^3y^2 - 24x^3y + 178x^3 - 2x^2y^3 + 16x^2y^2 - 60x^2y + 202x^2 - 12xy^3 + 14xy^2 - 56xy + 66x - 2y^5 + 4y^4 - 34y^3 + 22y^2 - 132y - 14$

$g = 2, i = 14, m = 2$, neighbours 2, 4, 6, 9, 24, 29

W $-4x^3 + 2x^2 + 2x$

A $2x^3y^2 + 8x^3y + 28x^3 - 2x^2y^2 - 4x^2y - 10x^2 - 2xy^3 - 8xy^2 - 16xy - 4x$

$$\begin{aligned}
& +2y^3 - 4y^2 + 10y \\
B \quad & -4x^5y^2 - 12x^5y - 52x^5 - 4x^3y^3 - 12x^3y^2 - 52x^3y + 4x^3 \\
& +4x^2y^4 + 20x^2y^3 + 58x^2y^2 + 12x^2y - 8x^2 - 2xy^5 - 8xy^4 - 20xy^3 \\
& -20xy^2 - 52xy - 6x - 2y^5 - 24y^4 - 22y^3 - 54y^2 + 16y + 6 \\
g = 2, i = 16, m = 2, \text{neighbours } 3, 4, 7, 9, 29 \\
W \quad & 4x^4 + 6x^3 - 6x^2 - 4x \\
A \quad & -8x^4 - 10x^3 + 6x^2 + 2xy - 8y \\
B \quad & -28x^5 + 28x^4 + 28x^3y + 80x^3 + 8x^2y^3 - 4x^2y^2 - 20x^2y - 34x^2 \\
& -2xy^3 - 8xy^2 - 16xy - 26x - 4y^4 + 8y^3 + 8y^2 + 32y + 4 \\
g = 2, i = 17, m = 88, \text{neighbours } 7, 9, 13, 28 \\
W \quad & 56x^5 - 72x^4 - 120x^3 + 40x^2 - 72x + 136 \\
A \quad & -1568x^6 + 4032x^5 + 4128x^4 + 1232x^3y^2 - 252x^3y - 16579x^3 + 784x^2y^3 \\
& +3136x^2y - 1520x^2 - 2016xy^3 - 308xy^2 - 8001xy + 14259x - 784y^4 + 336y^3 \\
& -13508y^2 + 11443y - 3008 \\
B \quad & -2464x^5y^2 + 504x^5y + 12966x^5 - 2464x^4y^2 + 504x^4y + 12966x^4 \\
& -2464x^3y^3 - 1344x^3y^2 + 13344x^3y - 24224x^3 + 1568x^2y^4 \\
& -9728x^2y^3 + 24944x^2y^2 - 5092x^2y + 10435x^2 - 1232xy^5 - 644xy^4 \\
& -205xy^3 + 22576xy^2 + 49604xy + 35958x - 784y^6 + 336y^5 - 16644y^4 + 19747y^3 \\
& -33472y^2 + 40008y - 23821 \\
g = 2, i = 19, m = 32, \text{neighbours } 9, 13, 20, 24, 38, 25, 39 \\
W \quad & -128x^3 + 192x^2 + 128x \\
A \quad & -1024x^8 + 512x^7y + 5120x^7 + 512x^6y^2 - 4096x^6y - 4096x^6 - 1024x^5y^2 \\
& +3072x^5y - 15872x^5 + 2048x^4y^3 + 1024x^4y^2 + 29440x^4y + 29696x^4 \\
& +256x^3y^4 - 128x^3y^3 + 6816x^3y^2 + 28248x^3y + 213978x^3 \\
& -256x^2y^5 + 3072x^2y^4 + 3584x^2y^3 + 41728x^2y^2 - 2560x^2y \\
& -38144x^2 + 512xy^5 + 384xy^4 + 5024xy^3 + 18968xy^2 + 85034xy - 415924x - 256y^6 \\
& +2048y^5 + 2688y^4 + 27424y^3 + 37304y^2 + 121474y - 190682 \\
B \quad & -1024x^5y^4 - 11008x^5y^3 - 53056x^5y^2 - 176816x^5y - 435124x^5 \\
& -1024x^4y^4 - 11008x^4y^3 - 53056x^4y^2 - 176816x^4y - 435124x^4 \\
& +1024x^3y^5 + 9216x^3y^4 + 33792x^3y^3 + 83968x^3y^2 + 125696x^3y \\
& +72704x^3 - 1536x^2y^6 - 5120x^2y^5 - 45568x^2y^4 - 86400x^2y^3 \\
& -181920x^2y^2 - 152152x^2y - 220634x^2 - 256xy^7 - 2432xy^6 - 7584xy^5 \\
& -52056xy^4 - 251226xy^3 - 315552xy^2 - 1058584xy - 426164x + 256y^8 - 1792y^7 \\
& -4480y^6 - 42912y^5 - 90968y^4 - 310490y^3 - 285184y^2 - 815552y - 186586 \\
g = 2, i = 20, m = 2, \text{neighbours } 4, 6, 19, 37 \\
W \quad & -4x^2 + 8x - 8 \\
A \quad & -2x^2 + 2x - 2y + 14 \\
B \quad & 4x^2y^2 + 8x^2y - 32x^2 - 4xy^2 + 30x - 2y^3 - 16y^2 + 28y - 78 \\
g = 2, i = 24, m = 4, \text{neighbours } 2, 9, 14, 19, 37 \\
W \quad & -8x^4 + 8x^2 \\
A \quad & -32x^4 + 16x^3y - 30x^3 + 8x^2y^2 + 32x^2y + 88x^2 - 8xy^2 - 34xy + 52x - 8y^3 \\
& +8y^2 - 38y - 54 \\
B \quad & -48x^5y + 44x^5 + 48x^4y - 44x^4 - 48x^3y^2 - 16x^3y + 32x^2y^3 + 96x^2y \\
& +14x^2 - 16xy^4 - 2xy^3 - 40xy^2 + 24xy - 56x - 8y^5 - 102y^3 + 72y^2 - 184y - 54 \\
g = 2, i = 25, m = 1, \text{neighbours } 2, 19, 28 \\
W \quad & 2x^2 - 4x \\
A \quad & 0 \\
B \quad & -x^2y - 5x^2 - 3xy + 5x - y^2 + 1 \\
g = 2, i = 26, m = 1, \text{neighbours } 7, 9, 27, 28 \\
W \quad & -x^3 - 3x^2 + x + 3 \\
A \quad & -x^2 - x - y + 2 \\
B \quad & -6x^3 + x^2y^2 - 2x^2 + 3xy^2 + 6xy + 7x + 2y^3 + 2y^2 + 6y + 5 \\
g = 2, i = 27, m = 1, \text{neighbours } 33, 36, 38, 26 \\
W \quad & -4x^2 - 4x \\
A \quad & x^2 + 2x + y - 1 \\
B \quad & -6x^3 - 10x^2 + 6xy - 15x - y^3 - 2y^2 + 2y - 5 \\
g = 2, i = 28, m = 2, \text{neighbours } 3, 12, 17, 25, 26 \\
W \quad & -4x^4 - 20x^3 - 28x^2 - 12x \\
A \quad & -8x^4 - 12x^3y - 39x^3 + 2x^2y^2 + 4x^2y - 318x^2 - 8xy^2 + 37xy - 567x + 2y^3 \\
& +38y^2 - 212y - 248
\end{aligned}$$

B	$16x^5y - 74x^5 - 32x^4y + 148x^4 - 16x^3y^2 + 88x^3y - 320x^3 + 8x^2y^3$ $-12x^2y^2 + 116x^2y - 173x^2 + 12xy^4 + 39xy^3 + 14xy^2 + 172xy + 327x - 2y^5$ $-40y^4 + 256y^3 - 166y^2 + 964y + 248$
$g = 2, i = 29, m = 2$, neighbours 31, 6, 14, 16, 37	
W	$-4x^3 + 2x^2 + 2x$
A	$-2x^3 + 2x^2y - 6x^2 - 2xy + 22x - 2y^2 + 8y + 10$
B	$8x^5 + 16x^4 + 8x^3y + 4x^3 - 4x^2y + 6x^2 + 2xy^3 - 4xy^2 + 2xy + 12x - 2y^4$ $+6y^3 - 12y^2 + 38y + 10$
$g = 2, i = 35, m = 1$, neighbours 6	
W	1
A	0
B	1
$g = 2, i = 36, m = 1$, neighbours 33, 0, 12, 27	
W	$2x^5 + 6x^3 + 4x$
A	$-2x^6 - 12x^4 + x^2y^3 + 4x^2y - 25x^2 - y^4 + 3y^3 - 2y^2 + 15y - 15$
B	$2x^5 + 2x^3y + 10x^3 + 2x^2y^4 + 6x^2y^3 + 4x^2y^2 + 18x^2y - 6x^2$ $-xy^3 + 2xy^2 + 15x - y^6 + 2y^5 - 4y^4 + 22y^3 + 62y + 15$
$g = 2, i = 37, m = 2$, neighbours 2, 20, 24, 29	
W	$-4x^3 + 4x^2$
A	$-2x^3 - 2x^2y - 10x^2 + 4xy + 14x + 2y^2 + 14y + 12$
B	$4x^5 + 4x^4 + 4x^3y + 20x^3 - 8x^2y - 6x^2 + 2xy^3 - 4xy^2 + 2xy + 10x + 2y^4$ $+16y^3 + 20y^2 + 62y + 16$
$g = 2, i = 38, m = 1$, neighbours 19, 27	
W	$-2x$
A	0
B	$x^2 + 3x + y - 1$
$g = 2, i = 39, m = 1$, neighbours 19	
W	-1
A	0
B	1

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 1\}, k_{ij} = 1$	
α	1
β	$-2x^3 + y^2 + 3$
γ	$2x^3 + 1$
$\{i, j\} = \{0, 9\}, k_{ij} = 2^2$	
α	1
β	$2x^3 + 2x^2 + 2x + y^2 + 7$
γ	$-2x^3 - 2x^2 - 2x - 3$
$\{i, j\} = \{0, 20\}, k_{ij} = 1$	
α	x
β	$-2x^3 - 2x^2 + xy^2 + 4x + 1$
γ	$-2x^3 - 2x^2 + 1$
$\{i, j\} = \{1, 7\}, k_{ij} = 1$	
α	0
β	-1
γ	1
$\{i, j\} = \{1, 11\}, k_{ij} = 1$	
α	-81
β	$-162x^3 + 162x^2y - 162xy^2 + 54y^3 - 18y^2 + 6y - 245$
γ	$-54y^3 - 9y^2 + 3y - 82$
$\{i, j\} = \{1, 22\}, k_{ij} = 1$	
α	$-x^4 + y + 1$
β	$-2x^7 + 2x^6y - x^5y + x^5 - 3x^4y^2 + x^4y - 3x^4 + x^3 - x^2y - xy^2 + xy - 4x$ $+y^3 + 3y^2 + 3y + 9$
γ	$2x^4y^2 - x^4y + x^4 - 2y^2 + y - 4$

$$\begin{aligned}
&\{i, j\} = \{1, 28\}, k_{ij} = 3 \\
&\alpha \quad 2x^4 - y + 3 \\
&\beta \quad 4x^7 - 4x^6y - 6x^5y - 2x^5 - 2x^4y^2 - 10x^4y - 2x^4 + 6x^3 - 6x^2y + xy^2 \\
&\quad - 8xy - y^3 - 2y^2 - 17y + 3 \\
&\gamma \quad -4x^4y^2 - 6x^4y - 2x^4 - 5y^2 - 8y \\
&\{i, j\} = \{2, 5\}, k_{ij} = 3^3 \\
&\alpha \quad -8 \\
&\beta \quad 8x^3 + 4x^2 + 2x - 8y^2 + 8y - 35 \\
&\gamma \quad -8x^3 - 4x^2 - 2x - 5 \\
&\{i, j\} = \{2, 12\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad -2 \\
&\beta \quad -2x^3 + 2x^2 - 2x - 2y^2 + 2y - 9 \\
&\gamma \quad 2x^3 - 2x^2 + 2x - 1 \\
&\{i, j\} = \{2, 14\}, k_{ij} = 5 \\
&\alpha \quad x^3 - 2 \\
&\beta \quad 2x^5 + x^3y^2 - x^3y + 3x^3 - 3x^2 - 2x - 2y^2 + 2y - 7 \\
&\gamma \quad -2x^5 + 2x^3 + 3x^2 + 2x - 3 \\
&\{i, j\} = \{2, 16\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad -x^3 - 3 \\
&\beta \quad 2x^5 - 2x^4 - x^3y^2 + x^3y - x^3 + x^2 - x - 3y^2 + 3y - 13 \\
&\gamma \quad 2x^5 - 2x^4 + 4x^3 + x^2 - x + 2 \\
&\{i, j\} = \{2, 23\}, k_{ij} = 3 \\
&\alpha \quad -x^3 - 2 \\
&\beta \quad -2x^5 - x^3y^2 + x^3y - 7x^3 - 5x^2 + 2x - 2y^2 + 2y - 15 \\
&\gamma \quad 2x^5 + 2x^3 + 5x^2 - 2x + 5 \\
&\{i, j\} = \{2, 25\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
&\alpha \quad 17x^3 - 120 \\
&\beta \quad -34x^5 - 68x^4 + 17x^3y^2 - 17x^3y - 51x^3 - 49x^2 - 30x - 120y^2 + 120y - 660 \\
&\gamma \quad -34x^5 - 68x^4 - 136x^3 - 49x^2 - 30x - 60 \\
&\{i, j\} = \{3, 4\}, k_{ij} = 3^2 \\
&\alpha \quad 1 \\
&\beta \quad 2x^3 + 2x^2 + 2x + y^2 + y + 8 \\
&\gamma \quad -2x^3 - 2x^2 - 2x - 3 \\
&\{i, j\} = \{3, 5\}, k_{ij} = 3^2 \cdot 5 \\
&\alpha \quad 8 \\
&\beta \quad -8x^3 + 4x^2 - 2x + 8y^2 + 8y + 37 \\
&\gamma \quad 8x^3 - 4x^2 + 2x + 3 \\
&\{i, j\} = \{3, 9\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad 1 \\
&\beta \quad 2x^3 + 4x^2 + 8x + y^2 + y + 22 \\
&\gamma \quad -2x^3 - 4x^2 - 8x - 17 \\
&\{i, j\} = \{3, 21\}, k_{ij} = 2^7 \\
&\alpha \quad 7x^3 - 11 \\
&\beta \quad -14x^5 + 14x^4 + 7x^3y^2 + 7x^3y + 49x^3 - 27x^2 - 43x - 11y^2 - 11y + 42 \\
&\gamma \quad -14x^5 + 14x^4 + 14x^3 - 27x^2 - 43x + 97 \\
&\{i, j\} = \{4, 5\}, k_{ij} = 3 \\
&\alpha \quad 3 \\
&\beta \quad -6x^3 - 6x^2y - 6xy^2 - 2y^3 + 2y^2 + 7 \\
&\gamma \quad 2y^3 + y^2 + 5 \\
&\{i, j\} = \{4, 7\}, k_{ij} = 3 \cdot 5 \\
&\alpha \quad 4 \\
&\beta \quad -8x^3 - 8x^2y - 8xy^2 - 4y^3 + 4y^2 + 2y + 7 \\
&\gamma \quad 4y^3 + 4y^2 + 2y + 11 \\
&\{i, j\} = \{4, 11\}, k_{ij} = 3^2 \\
&\alpha \quad 1 \\
&\beta \quad -2x^3 - 2x^2y - 2xy^2 + 2y^3 + 6y^2 + 6y + 15 \\
&\gamma \quad -2y^3 - 3y^2 - 3y - 8 \\
&\{i, j\} = \{4, 16\}, k_{ij} = 3^3 \cdot 5 \\
&\alpha \quad -14x^4 - 49y + 135
\end{aligned}$$

β	$28x^7 + 28x^6y + 42x^5y - 84x^5 + 14x^4y^2 - 42x^4y - 70x^4 - 256x^3$
	$-256x^2y - 49xy^2 - 279xy + 313x - 49y^3 - 72y^2 + 83y + 178$
γ	$28x^4y^2 - 42x^4y + 84x^4 - 207y^2 + 279y - 313$
$\{i, j\} = \{4, 19\}, k_{ij} = 3^3$	
α	$-2x^4 + 6y - 1$
β	$4x^7 + 4x^6y - 2x^5y - 4x^5 + 2x^4y^2 - 2x^4y + 6x^4 + 8x^3 + 8x^2y + 6xy^2$
	$-xy + 28x + 6y^3 + y^2 + 23y - 27$
γ	$4x^4y^2 + 2x^4y + 4x^4 + 2y^2 + y - 28$
$\{i, j\} = \{4, 20\}, k_{ij} = 5^2$	
α	$2x^4 - 4y + 1$
β	$-4x^7 - 4x^6y + 2x^5y - 4x^5 - 2x^4y^2 - 6x^4y + 2x^4 - 8x^3 - 8x^2y - 4xy^2$
	$+xy - 24x - 4y^3 - 3y^2 - 21y + 23$
γ	$-4x^4y^2 - 2x^4y + 4x^4 - 4y^2 - y + 24$
$\{i, j\} = \{4, 24\}, k_{ij} = 3^3$	
α	$-2x^4 + 6y - 1$
β	$4x^7 + 4x^6y - 2x^5y - 4x^5 - 6x^4y^2 - 6x^4y - 2x^4 + 8x^3 + 8x^2y + 6xy^2$
	$-xy + 28x + 6y^3 - 3y^2 + 21y + 29$
γ	$4x^4y^2 + 2x^4y + 4x^4 + 2y^2 + y - 28$
$\{i, j\} = \{5, 17\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5$	
α	$68x^4 - 289y + 540$
β	$68x^7 - 34x^6y - 170x^5y - 340x^5 + 102x^4y^2 + 340x^4y + 374x^4 + 574x^3$
	$-287x^2y + 578xy^2 - 211xy + 1601x - 289y^3 - 329y^2 - 734y - 1331$
γ	$-34x^4y^2 - 340x^4y - 680x^4 + 869y^2 - 422y + 3202$
$\{i, j\} = \{5, 18\}, k_{ij} = 2^5 \cdot 5^2$	
α	$-7182x^4 + 96824y + 800$
β	$20748x^7 - 1064x^6y - 7448x^5y - 44688x^5 - 6118x^4y^2 + 61446x^4y$
	$-20349x^4 + 134094x^3 - 126932x^2y + 141512xy^2 - 37324xy - 103677x + 13034y^3$
	$-20297y^2 + 287189y + 33608$
γ	$55860x^6 + 18620x^5y - 55860x^5 - 1064x^4y^2 + 22344x^4y - 33516x^4$
	$+83790x^2y^2 - 218785x^2y + 273770x^2 - 83790xy^2 + 99015xy - 273770x + 83790y^3$
	$+21097y^2 + 100107y + 66416$
$\{i, j\} = \{5, 21\}, k_{ij} = 2 \cdot 3^2 \cdot 7$	
α	$4x^4 - 9y$
β	$4x^7 - 2x^6y - 6x^5y + 16x^5 + 2x^4y^2 - 14x^4y + 18x^4 + 18x^3 - 9x^2y$
	$+18xy^2 - 27xy + 63x - 9y^3 + 27y^2 - 63y + 63$
γ	$-2x^4y^2 - 12x^4y + 32x^4 + 27y^2 - 54y + 126$
$\{i, j\} = \{5, 27\}, k_{ij} = 2^4 \cdot 3^3$	
α	$-4x^4 + 15y - 2$
β	$-4x^7 + 2x^6y + 8x^5y + 4x^5 - 2x^4y^2 + 14x^4y + 6x^4 - 8x^3 + 4x^2y$
	$-30xy^2 + 4xy - 217x + 15y^3 - 58y^2 + 124y - 435$
γ	$2x^4y^2 + 16x^4y + 8x^4 - 56y^2 + 8y - 434$
$\{i, j\} = \{5, 28\}, k_{ij} = 2^2 \cdot 5$	
α	$4x^4 + y + 2$
β	$4x^7 - 2x^6y - 8x^5y + 4x^5 + 2x^4y^2 - 18x^4y + 10x^4 + 8x^3 - 4x^2y$
	$-2xy^2 - 4xy + 9x + y^3 - 6y^2 - 12y + 19$
γ	$-2x^4y^2 - 16x^4y + 8x^4 - 8y^2 - 8y + 18$
$\{i, j\} = \{6, 10\}, k_{ij} = 2^2$	
α	2
β	$4x^3 - 4x^2y - 4x^2 + 4xy^2 + 8xy + 4x - 2y^3 - 3y^2 + 9$
γ	$2y^3 + 9y^2 + 12y + 11$
$\{i, j\} = \{6, 14\}, k_{ij} = 3^2$	
α	$2x^4 - y - 4$
β	$4x^7 - 4x^6y - 4x^6 - 6x^5y - 2x^5 + 6x^4y^2 + 8x^4y + 4x^4 - 10x^3 + 10x^2y$
	$+10x^2 + xy^2 + 13xy + 9x - y^3 - 14y^2 - 23y - 13$
γ	$-4x^4y^2 - 14x^4y - 6x^4 + 11y^2 + 33y + 19$
$\{i, j\} = \{6, 20\}, k_{ij} = 2^2$	
α	-1
β	$-2x^3 + 2x^2y + 2x^2 - 3xy + y^2 + 2y - 3$
γ	$2y^2 + y + 2$

$$\begin{aligned}
&\{i, j\} = \{6, 29\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
&\alpha \quad 2x^4 - 8y - 23 \\
&\beta \quad 4x^7 - 4x^6y - 4x^6 + 2x^5y + 6x^5 + 6x^4y^2 + 4x^4y - 8x^4 - 38x^3 + 38x^2y \\
&\quad + 38x^2 + 8xy^2 - 39xy + 11x - 8y^3 - 61y^2 - 54y - 132 \\
&\gamma \quad -4x^4y^2 - 6x^4y + 2x^4 + 46y^2 + 37y + 49 \\
&\{i, j\} = \{7, 9\}, k_{ij} = 2^3 \\
&\alpha \quad 2 \\
&\beta \quad 4x^3 - 4x^2y + 4x^2 + 4xy^2 - 8xy + 4x - 2y^3 + 5y^2 + 1 \\
&\gamma \quad 2y^3 - 7y^2 + 12y - 5 \\
&\{i, j\} = \{7, 10\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 2 \\
&\beta \quad 4x^3 - 4x^2y + 4x^2 + 4xy^2 - 8xy + 4x - 2y^3 + 7y^2 - 5y + 8 \\
&\gamma \quad 2y^3 - 5y^2 + 7y + 2 \\
&\{i, j\} = \{7, 13\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \\
&\alpha \quad 1 \\
&\beta \quad 2x^3 - 2x^2y + 2x^2 + 2xy^2 - 4xy + 2x + 2y^3 + 8y^2 + 26y + 121 \\
&\gamma \quad -2y^3 - y^2 - 16y - 59 \\
&\{i, j\} = \{7, 16\}, k_{ij} = 2 \cdot 5^2 \\
&\alpha \quad 2x^4 - 5y + 8 \\
&\beta \quad 4x^7 - 4x^6y + 4x^6 - 2x^5y - 2x^4y^2 + 4x^4y - 2x^4 + 16x^3 - 16x^2y \\
&\quad + 16x^2 + 5xy^2 + 2xy + 25x - 5y^3 - 8y^2 + 6y + 42 \\
&\gamma \quad -4x^4y^2 + 6x^4y - 4x^4 - 11y^2 + 34y + 9 \\
&\{i, j\} = \{7, 21\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad x^4 - 2y + 2 \\
&\beta \quad 2x^7 - 2x^6y + 2x^6 - x^5y + 2x^5 - x^4y^2 + 3x^4 + 8x^3 - 8x^2y + 8x^2 \\
&\quad + 2xy^2 + 2xy + 20x - 2y^3 - 6y^2 - 2y + 34 \\
&\gamma \quad -2x^4y^2 + 3x^4y - 6y^2 + 18y + 12 \\
&\{i, j\} = \{7, 26\}, k_{ij} = 2^3 \cdot 3^2 \\
&\alpha \quad 2x^4 - 7y + 16 \\
&\beta \quad 4x^7 - 4x^6y + 4x^6 - 2x^5y - 8x^5 - 2x^4y^2 + 12x^4y - 18x^4 + 40x^3 \\
&\quad - 40x^2y + 40x^2 + 7xy^2 - 2xy - 23x - 7y^3 - 24y^2 + 92y - 70 \\
&\gamma \quad -4x^4y^2 + 6x^4y - 12x^4 - 33y^2 + 78y - 63 \\
&\{i, j\} = \{8, 29\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad -x^3 - 3 \\
&\beta \quad -2x^5 - 4x^4 - x^3y^2 + 2x^3y - 14x^3 - 15x^2 - 9x - 3y^2 + 6y - 27 \\
&\gamma \quad 2x^5 + 4x^4 + 6x^3 + 15x^2 + 9x + 3 \\
&\{i, j\} = \{9, 14\}, k_{ij} = 5 \cdot 7 \\
&\alpha \quad 6x^4 - 9y - 4 \\
&\beta \quad -12x^7 - 12x^6y - 12x^6 + 18x^5y + 6x^5 + 18x^4y^2 + 24x^4y + 2x^3 + 2x^2y \\
&\quad + 2x^2 - 9xy^2 + 3xy - 43x - 9y^3 - 6y^2 - 31y - 39 \\
&\gamma \quad -12x^4y^2 - 42x^4y - 18x^4 + 11y^2 + y + 45 \\
&\{i, j\} = \{9, 16\}, k_{ij} = 2 \cdot 7 \\
&\alpha \quad 55x^4 + 655y + 369 \\
&\beta \quad -155x^7 - 200x^6y - 200x^6 - 300x^5y + 200x^5 - 145x^4y^2 + 90x^4y + 545x^4 \\
&\quad + 886x^3 + 5080x^2y + 5080x^2 + 250xy^2 + 6820xy + 4432x + 520y^3 + 5309y^2 + 5484y \\
&\quad + 355 \\
&\gamma \quad 45x^6 + 45x^5y - 200x^4y^2 - 190x^4y - 400x^4 + 135x^2y^2 - 495x^2y \\
&\quad - 1134x^2 - 135xy^2 - 3699xy - 3060x - 135y^3 + 5460y^2 + 8173y + 3708 \\
&\{i, j\} = \{9, 19\}, k_{ij} = 2 \cdot 7^2 \\
&\alpha \quad -x^4 + 5y + 10 \\
&\beta \quad 2x^7 + 2x^6y + 2x^6 - x^5y - 2x^5 + x^4y^2 + 2x^4y + 3x^4 - 12x^3 - 12x^2y \\
&\quad - 12x^2 + 5xy^2 + 54x + 5y^3 - 12y^2 + 25y - 22 \\
&\gamma \quad 2x^4y^2 + 5x^4y + 4x^4 - 17y^2 - 24y - 66 \\
&\{i, j\} = \{9, 21\}, k_{ij} = 2^4 \cdot 3^2 \\
&\alpha \quad x^4 - y - 37 \\
&\beta \quad -2x^7 - 2x^6y - 2x^6 - 3x^5y - x^4y^2 - 2x^4y + x^4 + 76x^3 + 76x^2y + 76x^2 \\
&\quad - xy^2 + 119xy - 22x - y^3 + 41y^2 + 65y - 83 \\
&\gamma \quad -2x^4y^2 - x^4y - 2x^4 + 77y^2 + 33y + 98 \\
&\{i, j\} = \{9, 26\}, k_{ij} = 2^6 \cdot 3 \cdot 5 \cdot 7
\end{aligned}$$

α	$-38x^4 - 361y + 1718$
β	$76x^7 + 76x^6y + 76x^6 + 114x^5y - 228x^5 + 38x^4y^2 - 152x^4y - 494x^4$ $-3284x^3 - 3284x^2y - 3284x^2 - 361xy^2 - 3406xy + 1967x - 361y^3 - 844y^2 + 2084y$ $+5500$
γ	$76x^4y^2 + 38x^4y + 304x^4 - 2923y^2 - 3162y - 5251$
$\{i, j\} = \{10, 15\}, k_{ij} = 2 \cdot 3^2$	
α	$2x^4 - 4y + 7$
β	$-4x^7 - 4x^6y + 4x^6 - 2x^5y + 10x^5 - 2x^4y^2 + 12x^4y - 12x^4 - 18x^3$ $-18x^2y + 18x^2 - 4xy^2 - 15xy + 19x - 4y^3 - 11y^2 + 38y - 26$
γ	$-4x^4y^2 + 10x^4y - 14x^4 - 14y^2 + 51y - 37$
$\{i, j\} = \{11, 18\}, k_{ij} = 2 \cdot 3$	
α	$4x^4 - y + 5$
β	$-4x^7 - 2x^6y - 2x^6 + 10x^5y + 2x^5 + 6x^4y^2 + 18x^4y + 2x^4 - 6x^3$ $-3x^2y - 3x^2 - 2xy^2 + 13xy - y^3 + 5y^2 + 23y - 1$
γ	$-2x^4y^2 - 24x^4y - 6x^4 + y^2 - 32y - 3$
$\{i, j\} = \{11, 24\}, k_{ij} = 2^3 \cdot 3^2$	
α	$26x^4 - 18$
β	$52x^7 + 26x^4y^2 + 234x^4y + 273x^4 - 88x^3 + 293x - 52y^3 + 47y^2 + 173y + 218$
γ	$-156x^6 + 52x^5y + 208x^5 - 208x^4y - 364x^4 + 52x^2y^2 - 169x^2y + 95x^2$ $-52xy^2 + 81xy - 183x + 52y^3 - 65y^2 - 155y - 308$
$\{i, j\} = \{11, 28\}, k_{ij} = 2^{11}$	
α	$2772x^4 + 88641y + 88129$
β	$-9828x^7 + 2142x^6y + 2142x^6 + 25704x^5 + 4914x^4y^2 + 22680x^4y + 11340x^4$ $+420120x^3 - 835516x^2y - 835516x^2 + 134946xy^2 - 1231650xy - 3112453x + 81585y^3$ $-890712y^2 - 2587856y - 1794505$
γ	$14112x^6 - 14112x^5y - 42336x^5 + 2142x^4y^2 + 18396x^4y + 21294x^4$ $+7056x^2y^2 - 194040x^2y - 1029224x^2 - 14112xy^2 + 2479232xy + 4149600x$ $-7056y^3 - 890200y^2 - 2942932y - 1880586$
$\{i, j\} = \{12, 16\}, k_{ij} = 2$	
α	$-4x^4 - 41y + 11$
β	$-20x^7 + 14x^6y - 14x^6 - 14x^5y - 42x^5 + 10x^4y^2 - 14x^4y - 26x^4$ $+442x^3 - 175x^2y + 175x^2 + 14xy^2 + 161xy - 266x - 17y^3 - 221y^2 - 285y - 9$
γ	$48x^6 - 48x^5y + 14x^4y^2 + 98x^4 - 24x^2y^2 - 12x^2y - 876x^2 + 24xy^2$ $+804xy + 84x + 24y^3 - 215y^2 + 100y + 273$
$\{i, j\} = \{12, 20\}, k_{ij} = 2^3 \cdot 5$	
α	$-4x^4 + y + 17$
β	$4x^7 + 2x^6y - 2x^6 - 6x^5y + 2x^5 - 2x^4y^2 + 8x^4y - 14x^3 - 7x^2y + 7x^2$ $+2xy^2 + 27xy - 12x + y^3 + 7y^2 - 40y + 6$
γ	$2x^4y^2 + 8x^4y - 2x^4 - 11y^2 - 40y + 17$
$\{i, j\} = \{12, 21\}, k_{ij} = 2^4 \cdot 3 \cdot 7$	
α	$4x^4 - 5y - 121$
β	$-4x^7 - 2x^6y + 2x^6 + 2x^5y + 2x^5 + 2x^4y^2 + 122x^3 + 61x^2y - 61x^2$ $-10xy^2 - 19xy - 244x - 5y^3 - 45y^2 - 68y - 92$
γ	$-2x^4y^2 - 6x^4 + 81y^2 - 84y + 549$
$\{i, j\} = \{13, 17\}, k_{ij} = 2^4 \cdot 7$	
α	$4x^4 + 7y + 10$
β	$4x^7 - 2x^6y - 4x^6 - 10x^5y + 16x^5 + 6x^4y^2 + 16x^4y - 26x^4 + 38x^3$ $-19x^2y - 38x^2 - 14xy^2 - 11xy + 47x + 7y^3 + 43y^2 + 40y - 51$
γ	$-2x^4y^2 - 28x^4y + 24x^4 - 47y^2 - 98y + 18$
$\{i, j\} = \{13, 19\}, k_{ij} = 2 \cdot 3 \cdot 5^3$	
α	$-7412x^4 - 166879y - 67754$
β	$-17876x^7 - 1526x^6y - 3052x^6 - 1526x^5y + 33572x^5 - 8938x^4y^2$ $-31610x^4y + 5450x^4 - 158214x^3 + 698019x^2y + 1396038x^2 + 166334xy^2$ $-898177xy - 1144809x - 125023y^3 + 768259y^2 + 225273y - 1338025$
γ	$-20928x^6 - 20928x^5y - 62784x^5 - 1526x^4y^2 - 9156x^4y - 22672x^4$ $-41856x^2y^2 + 319152x^2y - 185280x^2 - 41856xy^2 + 1556976xy + 2290368x$ $+41856y^3 + 627823y^2 + 3057130y + 2607546$
$\{i, j\} = \{14, 22\}, k_{ij} = 2^2$	
α	$-2y + 4$
β	$x^7 - 2x^6 - 2x^4y - 2x^2y^2 + 4x^2y - 8x^2 - 7x - 2y^3 + 4y^2 - 7y + 12$

$$\begin{aligned}
&\gamma \quad -x^7 + 7x - y + 2 \\
&\{i, j\} = \{14, 29\}, k_{ij} = 3 \cdot 5 \cdot 7 \\
&\alpha \quad -32y + 72 \\
&\beta \quad 16x^7 - 32x^6 - 16x^5 - 40x^4y - 8x^3 - 32x^2y^2 + 72x^2y - 128x^2 - 130x \\
&\quad -32y^3 + 72y^2 - 108y + 309 \\
&\gamma \quad -16x^7 + 8x^4y + 8x^3 + 138x - 20y - 33 \\
&\{i, j\} = \{15, 27\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
&\alpha \quad -2y + 12 \\
&\beta \quad -x^7 - 2x^6 + 2x^4y - 2x^4 + 18x^3 - 2x^2y^2 + 8x^2y + 10x^2 - 29x + 2y^3 \\
&\quad -10y^2 - 3y - 42 \\
&\gamma \quad x^7 - 18x^3 + 29x + y - 6 \\
&\{i, j\} = \{15, 28\}, k_{ij} = 2^2 \cdot 5 \\
&\alpha \quad -2y + 10 \\
&\beta \quad -x^7 - 2x^6 - 2x^5 + x^4y - x^4 + 13x^3 - 2x^2y^2 + 6x^2y + 6x^2 + 16x + 2y^3 \\
&\quad -8y^2 + 12y - 12 \\
&\gamma \quad x^7 + x^4y - x^4 - 13x^3 + 10x - 12y - 2 \\
&\{i, j\} = \{16, 27\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad y \\
&\beta \quad x^7 + x^6 - 2x^5 - 4x^4y + 4x^4 + 8x^3 + x^2y^2 - x^2y + 6x^2 - xy^2 + xy - 9x \\
&\quad -3y^3 + 5y^2 - 27y + 43 \\
&\gamma \quad -x^7 + 3x^4y - 2x^4 - 8x^3 - 5x + 2y^3 - 3y^2 + 20y - 24 \\
&\{i, j\} = \{16, 29\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5 \cdot 7 \\
&\alpha \quad -3492x^4 + 1649y + 5950 \\
&\beta \quad 2619x^7 - 873x^6 - 5238x^5y - 1746x^5 + 1746x^4y^2 + 4656x^4y + 2328x^4 \\
&\quad -3056x^3 - 4947x^2y^2 + 9215x^2y + 178x^2 + 4947xy^2 - 3103xy + 10609x + 1649y^3 \\
&\quad -8407y^2 - 4713y - 21941 \\
&\gamma \quad -2619x^7 - 1746x^4y^2 + 1455x^4y - 582x^4 + 3056x^3 - 7731x + 3298y^3 \\
&\quad -5755y^2 + 7638y - 4130 \\
&\{i, j\} = \{17, 28\}, k_{ij} = 2^{11} \cdot 3 \\
&\alpha \quad 3276x^4 - 2093y + 680 \\
&\beta \quad -2457x^7 - 13923x^6 - 4914x^5y - 16380x^5 + 1638x^4y^2 - 7098x^4y - 9828x^4 \\
&\quad -1966x^3 + 6279x^2y^2 - 455x^2y - 3982x^2 + 6279xy^2 - 4387xy - 193x + 2093y^3 \\
&\quad +5345y^2 - 8131y + 5853 \\
&\gamma \quad 2457x^7 - 1638x^4y^2 - 11739x^4y - 6552x^4 + 1966x^3 - 5755x + 4186y^3 \\
&\quad +479y^2 - 238y + 5464 \\
&\{i, j\} = \{18, 24\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{19, 25\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
&\alpha \quad y + 2 \\
&\beta \quad -x^7 + 4x^6 - 4x^5 + 2x^4y + 2x^4 - 26x^3 + 4x^2y^2 - 12x^2y - 100x^2 \\
&\quad +4xy^2 - 12xy + 269x - 4y^2 - 153y - 166 \\
&\gamma \quad x^7 - 3x^6 - 3x^4y + x^4 + 26x^3 - 3x^2y^2 + 12x^2y + 123x^2 - 6xy^2 + 24xy \\
&\quad -175x - y^3 + 3y^2 + 142y - 138 \\
&\{i, j\} = \{19, 27\}, k_{ij} = 2 \cdot 3^2 \cdot 7 \\
&\alpha \quad -3y + 24 \\
&\beta \quad -x^7 - 3x^6 - 3x^5 + x^4y - 3x^4 + 31x^3 - 3x^2y^2 + 15x^2y + 39x^2 - 3xy^2 \\
&\quad +15xy - 21x + y^3 - 8y^2 + y - 73 \\
&\gamma \quad x^7 + 2x^4y - 3x^4 - 31x^3 + 60x + 2y^3 - 13y^2 - 10y - 29 \\
&\{i, j\} = \{20, 25\}, k_{ij} = 2^2 \\
&\alpha \quad y \\
&\beta \quad -x^7 + x^6 + x^5 - 3x^4y - x^4 - 5x^3 + x^2y^2 - x^2y + 5x^2 + xy^2 - xy + x \\
&\quad -3y^3 + 2y^2 - 15y + 3 \\
&\gamma \quad x^7 + 2x^4y + x^4 + 5x^3 + 4x + 2y^3 - y^2 + 10y + 1 \\
&\{i, j\} = \{22, 23\}, k_{ij} = 2^2 \cdot 5 \\
&\alpha \quad -2y - 22 \\
&\beta \quad -x^7 - 2x^6 - 2x^5 + x^4y + 10x^3 - 2x^2y^2 - 6x^2y + 56x^2 - 4xy^2 - 12xy \\
&\quad -124x + 2y^3 + 6y^2 - 47y - 227 \\
&\gamma \quad x^7 - 3x^4y - 4x^4 - 10x^3 + 216x - 4y^3 - 20y^2 + 79y - 21
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{23, 24\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad -y - 5 \\
&\beta \quad -x^7 - x^6 - x^5 - 4x^3 - x^2y^2 - x^2y + 4x^2 - 40x - y^3 - 3y^2 + 5y + 11 \\
&\gamma \quad x^7 - x^4y - 2x^4 + 4x^3 + 36x - 3y - 39 \\
&\{i, j\} = \{24, 29\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad -3y - 12 \\
&\beta \quad x^7 - 3x^6 + 3x^5 - x^4y - 3x^4 + 5x^3 - 3x^2y^2 - 3x^2y + 3x^2 + 3xy^2 + 3xy \\
&\quad + 15x - y^3 - 4y^2 - y - 29 \\
&\gamma \quad -x^7 - 2x^4y - 3x^4 - 5x^3 - 18x - 2y^3 - 5y^2 - 2y - 19 \\
&\{i, j\} = \{26, 27\}, k_{ij} = 2^3 \cdot 3^3 \\
&\alpha \quad -y + 14 \\
&\beta \quad -x^7 - x^6 + 2x^4y - 4x^4 + 90x^3 - x^2y^2 + 7x^2y + 56x^2 + xy^2 - 7xy - 157x \\
&\quad + 3y^3 - 29y^2 - 23y - 89 \\
&\gamma \quad x^7 - x^4y + 2x^4 - 90x^3 + 191x - 2y^3 + 19y^2 - 12y - 24 \\
&\{i, j\} = \{26, 28\}, k_{ij} = 2^8 \cdot 5 \\
&\alpha \quad -y + 12 \\
&\beta \quad -x^7 - x^6 - 2x^5 + 4x^4 + 40x^3 - x^2y^2 + x^2y + 28x^2 + xy^2 - xy + 235x \\
&\quad + 3y^3 - 11y^2 + 43y - 387 \\
&\gamma \quad x^7 + x^4y - 4x^4 - 40x^3 - 143x - 2y^3 + 7y^2 - 68y + 494 \\
&\{i, j\} = \{27, 28\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{0, 2\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{0, 7\}, k_{ij} = 2^2 \\
&\alpha \quad 1 \\
&\beta \quad -2x^3 - 2x^2 - 2x + y^2 + 1 \\
&\gamma \quad 2x^3 + 2x^2 + 2x + 3 \\
&\{i, j\} = \{0, 12\}, k_{ij} = 5 \\
&\alpha \quad 8 \\
&\beta \quad 8x^3 - 4x^2 + 2x + 8y^2 + 35 \\
&\gamma \quad -8x^3 + 4x^2 - 2x - 3 \\
&\{i, j\} = \{0, 13\}, k_{ij} = 2^2 \\
&\alpha \quad 2 \\
&\beta \quad -2x^3 + 2x^2 - 2x + 2y^2 + 9 \\
&\gamma \quad 2x^3 - 2x^2 + 2x - 1 \\
&\{i, j\} = \{0, 36\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad x^3 + 3 \\
&\beta \quad 2x^5 + x^3y^2 + 2x^3 + 7x^2 + 3x + 3y^2 + 5 \\
&\gamma \quad -2x^5 + 2x^3 - 7x^2 - 3x + 7 \\
&\{i, j\} = \{2, 3\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{2, 8\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{2, 9\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -2x^3 - y^2 + y - 6 \\
&\gamma \quad 2x^3 + 1 \\
&\{i, j\} = \{2, 24\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad x^3 - 3 \\
&\beta \quad 2x^5 - 2x^4 + x^3y^2 - x^3y + 7x^3 - 7x^2 + 3x - 3y^2 + 3y - 18 \\
&\gamma \quad -2x^5 + 2x^4 - 2x^3 + 7x^2 - 3x + 3
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{2, 37\}, k_{ij} = 2^2 \\
&\alpha \quad -3x^3 - 1 \\
&\beta \quad -6x^5 - 6x^4 - 3x^3y^2 + 3x^3y - 21x^3 - 11x^2 + x - y^2 + y - 4 \\
&\gamma \quad 6x^5 + 6x^4 + 6x^3 + 11x^2 - x - 1 \\
&\{i, j\} = \{3, 11\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad 2 \\
&\beta \quad 2x^3 + 2x^2 + 2x + 2y^2 + 2y + 13 \\
&\gamma \quad -2x^3 - 2x^2 - 2x - 3 \\
&\{i, j\} = \{3, 12\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 2 \\
&\beta \quad 2x^3 + 2y^2 + 2y + 11 \\
&\gamma \quad -2x^3 - 1 \\
&\{i, j\} = \{3, 16\}, k_{ij} = 3^2 \\
&\alpha \quad -x + 1 \\
&\beta \quad 2x^3 - 4x^2 - xy^2 - xy - 3x + y^2 + y + 8 \\
&\gamma \quad 2x^3 - 4x^2 + 2x + 3 \\
&\{i, j\} = \{3, 28\}, k_{ij} = 2^2 \cdot 5 \\
&\alpha \quad -x^3 - 4 \\
&\beta \quad 2x^5 - 4x^4 - x^3y^2 - x^3y - x^3 + 9x^2 - 20x - 4y^2 - 4y + 2 \\
&\gamma \quad 2x^5 - 4x^4 + 4x^3 + 9x^2 - 20x + 22 \\
&\{i, j\} = \{4, 14\}, k_{ij} = 3^3 \\
&\alpha \quad 8x^4 - 32y + 27 \\
&\beta \quad -16x^7 - 16x^6y + 24x^5y + 24x^5 + 24x^4y^2 + 24x^4y - 8x^4 - 62x^3 - 62x^2y \\
&\quad -32xy^2 + 57xy - 103x - 32y^3 + 57y^2 - 71y - 27 \\
&\gamma \quad -16x^4y^2 - 24x^4y - 24x^4 - 30y^2 - 57y + 103 \\
&\{i, j\} = \{6, 7\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{6, 13\}, k_{ij} = 2 \\
&\alpha \quad 1 \\
&\beta \quad 2x^3 - 2x^2y - 2x^2 + 2xy^2 + 4xy + 2x + 2y^3 + 12y^2 + 18y + 21 \\
&\gamma \quad -2y^3 - 9y^2 - 12y - 11 \\
&\{i, j\} = \{6, 35\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{7, 17\}, k_{ij} = 2^3 \cdot 3^2 \\
&\alpha \quad 2x^4 - 7y + 16 \\
&\beta \quad 4x^7 - 4x^6y + 4x^6 - 2x^5y - 8x^5 + 6x^4y^2 + 6x^4 + 40x^3 - 40x^2y + 40x^2 \\
&\quad + 7xy^2 - 2xy - 23x - 7y^3 + 42y^2 - 64y + 56 \\
&\gamma \quad -4x^4y^2 + 6x^4y - 12x^4 - 33y^2 + 78y - 63 \\
&\{i, j\} = \{9, 12\}, k_{ij} = 2^3 \\
&\alpha \quad -1 \\
&\beta \quad 2x^3 + 2x^2y + 2x^2 + 2xy^2 + 4xy + 2x - 2y^3 - 5 \\
&\gamma \quad 2y^3 + 3y^2 + 3y + 4 \\
&\{i, j\} = \{9, 17\}, k_{ij} = 2^4 \cdot 5 \\
&\alpha \quad 26x^4 - 169y - 6 \\
&\beta \quad -52x^7 - 52x^6y - 52x^6 + 130x^5y + 260x^5 + 78x^4y^2 + 156x^4y - 78x^4 \\
&\quad - 92x^3 - 92x^2y - 92x^2 - 169xy^2 + 178xy - 385x - 169y^3 + 86y^2 - 400y - 86 \\
&\gamma \quad -52x^4y^2 - 234x^4y - 312x^4 + 77y^2 - 362y + 293 \\
&\{i, j\} = \{9, 24\}, k_{ij} = 2^3 \\
&\alpha \quad x^4 - 4y - 6 \\
&\beta \quad -2x^7 - 2x^6y - 2x^6 + x^5y - 2x^5 + 3x^4y^2 + 4x^4y - 3x^4 + 8x^3 + 8x^2y \\
&\quad + 8x^2 - 4xy^2 + 2xy - 20x - 4y^3 - 14y^2 - 28y - 42 \\
&\gamma \quad -2x^4y^2 - 5x^4y + 12y^2 + 14y + 28 \\
&\{i, j\} = \{11, 12\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1
\end{aligned}$$

γ	1
$\{i, j\} = \{12, 28\}, k_{ij} = 2^2$	
α	$-136x^4 - 3281y - 643$
β	$-68x^7 + 170x^6y - 170x^6 - 1360x^5 + 34x^4y^2 - 340x^4y - 2346x^4$ $+56524x^3 - 19000x^2y + 19000x^2 - 2890xy^2 + 16830xy - 131283x - 2669y^3$ $-19490y^2 - 116332y - 94233$
γ	$408x^6 - 408x^5y - 408x^5 + 170x^4y^2 + 68x^4y + 3298x^4 - 612x^2y^2$ $+1734x^2y - 112578x^2 + 1224xy^2 + 91056xy + 130632x + 612y^3 - 16790y^2 - 66928y$ $+94880$
$\{i, j\} = \{12, 36\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5$	
α	$-290x^4 + 1131y - 1011$
β	$-928x^7 - 58x^6y + 58x^6 + 348x^5y + 1740x^5 - 232x^4y^2 + 1508x^4y$ $-1131x^4 + 4782x^3 - 4434x^2y + 4434x^2 - 5046xy^2 + 678xy + 525x - 2523y^3$ $+9687y^2 - 14292y + 7068$
γ	$2436x^6 - 812x^5y + 812x^5 - 58x^4y^2 + 1856x^4y - 5974x^4 + 3654x^2y^2$ $-11571x^2y - 147x^2 + 13650xy - 13650x + 3654y^3 - 9567y^2 + 18936y - 5337$
$\{i, j\} = \{14, 24\}, k_{ij} = 2 \cdot 3^2$	
α	$-y + 3$
β	$-x^7 - x^6 + x^5 - 2x^4y + 2x^3 - x^2y^2 + 3x^2y - 4x^2 + 10x - y^3 + 3y^2 - 3y$ $+27$
γ	$x^7 + x^4y - 2x^3 - 12x - y - 15$
$\{i, j\} = \{19, 20\}, k_{ij} = 2$	
α	0
β	-1
γ	1
$\{i, j\} = \{19, 24\}, k_{ij} = 2 \cdot 3^2$	
α	$4y - 2$
β	$2x^3y - x^3 - 4x^2y + 2x^2 - 4xy^2 - 2xy - 10x - 2y^3 + y^2 - 12y - 12$
γ	$-2x^3y + x^3 + 12x + 2y^3 - y^2 + 4y - 8$
$\{i, j\} = \{19, 38\}, k_{ij} = 2 \cdot 3 \cdot 7$	
α	$-y + 8$
β	$x^7 - x^6 - x^5 + 3x^4y - 5x^4 - 31x^3 - x^2y^2 + 5x^2y + 13x^2 - xy^2 + 5xy$ $+73x + 3y^3 - 20y^2 - 13y - 63$
γ	$-x^7 - 2x^4y + 3x^4 + 31x^3 - 60x - 2y^3 + 13y^2 + 10y + 29$
$\{i, j\} = \{19, 39\}, k_{ij} = 1$	
α	0
β	1
γ	-1
$\{i, j\} = \{20, 37\}, k_{ij} = 2^3$	
α	$-6x^3 - 2$
β	$12x^5 + 3x^3y^2 - 9x^3y + 15x^3 + 22x^2 - 12x + y^2 - 12y - 7$
γ	$12x^4 - 3x^3y^2 - 3x^3y - 15x^3 + 10x - y^2 - 10y - 17$
$\{i, j\} = \{25, 28\}, k_{ij} = 2^4 \cdot 7$	
α	$4y - 24$
β	$x^7 + 4x^6 + 8x^5 + 8x^4 - 10x^3 + 4x^2y^2 - 16x^2y - 4x^2 + 8xy^2 - 32xy + 29x$ $+8y^2 - 31y + 60$
γ	$-x^7 - 4x^4y + 4x^4 + 10x^3 - 37x - 4y^3 + 20y^2 - 13y + 76$
$\{i, j\} = \{36, 27\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5$	
α	$4x^4 - 24y + 54$
β	$-2x^7 - 12x^6 - 4x^5y + 4x^5 + 2x^4y^2 - 16x^4y + 16x^4 + 33x^3 - 24x^2y^2$ $+72x^2y - 66x^2 + 66xy - 48x - 24y^3 + 63y^2 - 102y + 114$
γ	$2x^7 - 2x^4y^2 + 4x^4y - 4x^4 - 33x^3 - 18x + 33y^2 - 36y - 12$
$\{i, j\} = \{37, 24\}, k_{ij} = 2^3 \cdot 3$	
α	$-2y - 8$
β	$-x^7 - 2x^6 - 2x^5 - x^4 - 5x^3 - 2x^2y^2 - 2x^2y + 2x^2 - 2xy^2 - 2xy - 16x$ $-y^2 - 2y - 21$
γ	$x^7 - 2x^4y - 3x^4 + 5x^3 + 18x - 2y^3 - 5y^2 - 11$
$\{i, j\} = \{37, 29\}, k_{ij} = 2^2 \cdot 3$	
α	$-y - 4$

β	$x^7 - x^6 - x^5 - 3x^4y - 5x^4 + 5x^3 - x^2y^2 - x^2y + x^2 - xy^2 - xy + 19x$
	$-3y^3 - 8y^2 - y - 27$
γ	$-x^7 + 2x^4y + 3x^4 - 5x^3 - 18x + 2y^3 + 5y^2 + 11$
	$\{i, j\} = \{38, 27\}, k_{ij} = 2^3 \cdot 3$
α	4
β	$-8x^2 - xy^2 - 3xy - 11x - 2y^2 - 6y - 5$
γ	$xy^2 + 3xy - 5x - 2y^2 + 2y - 15$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 7$ and for $x \leq -6$.

i	k_i	$\varphi_i(x)$	$x \geq 7$	$x \leq -6$
0	-1	$-2x^4 - x - 1$	+	+
1	-1	$-2x^4 + x^3 + 3x - 1$	+	+
2	-1	$-2x^4 - x + 4$	+	+
3	-1	$-2x^4 - x - 6$	+	+
4	-1	$-2x^4 - x^3 - 5x - 1$	+	+
5	-1	$-2x^4 + 8x^3 + 7x - 1$	+	+
6	-1	$-2x^4 + x^3 + 3x^2 + 6x + 4$	+	+
7	-1	$-2x^4 + x^3 - 3x^2 + 6x - 6$	+	+
8	-1	$-2x^4 - x + 15$	+	+
9	-1	$-2x^4 - x^3 + 3x^2 - 8x + 4$	+	+
10	-1	$-2x^4 - x^3 - 3x^2 - 8x - 6$	+	+
11	-1	$-2x^4 - 8x^3 + 12x^2 - 15x + 4$	+	+
12	-1	$-2x^4 - 8x^3 - 12x^2 - 15x - 6$	+	+
13	-1	$-2x^4 + 8x^3 + 24x^2 + 31x + 15$	+	+
14	-1	$-x^6 - 2x^4 - 4x^2 - x - 1$	+	+
15	1	$-x^6 + 5x^4 - 7x^2 + x + 6$	+	+
16	1	$-x^6 - 3x^5 - x^4 - x^3 - 4x^2 - 3x + 1$	+	+
17	-1	$-x^6 + 3x^5 - 5x^4 + x^3 - 4x^2 + 3x - 1$	+	+
18	-1	$-x^6 - 3x^5 - 5x^4 - x^3 - 4x^2 - 5x - 1$	+	+
19	1	$-x^6 + 3x^5 + 2x^4 - 5x^3 - 4x^2 + 8x + 6$	+	+
20	1	$-x^6 + 3x^5 - 4x^4 + 7x^3 - 10x^2 + 8x - 4$	+	+
21	1	$-x^6 - 3x^5 - 4x^4 - 7x^3 - 10x^2 - 6x - 4$	+	+
22	-1	$-x^6 + 6x^5 - 14x^4 + 8x^3 - 4x^2 + 7x - 1$	+	+
23	-1	$-x^6 + 4x^4 - 16x^2 - x + 15$	+	+
24	-1	$-x^6 - 3x^5 - 2x^4 + 5x^3 - 4x^2 - 8x + 4$	+	+
25	1	$-x^6 + 6x^5 - 13x^4 + 20x^3 - 19x^2 + 15x - 4$	+	+
26	1	$-x^6 - 3x^5 + 5x^4 + 11x^3 - 10x^2 - 15x + 17$	+	+
27	1	$-x^6 - 6x^5 - 7x^4 + 4x^3 + 5x^2 - 13x + 6$	+	+
28	1	$-x^6 - 6x^5 - 13x^4 - 20x^3 - 19x^2 - 13x - 4$	+	+
29	-1	$-x^6 + 6x^5 - 11x^4 - 4x^3 + 5x^2 + 13x + 4$	+	+
35	-1	$-2x^4 + x^3 + 6x^2 + 15x + 15$	+	+
36	-1	$-x^6 - 5x^4 - 7x^2 - x - 6$	+	+
37	-1	$-x^6 + 3x^5 - 2x^4 - 5x^3 - 4x^2 + 6x + 4$	+	+
38	1	$-x^6 + 6x^5 - 7x^4 - 4x^3 + 5x^2 + 15x + 6$	+	+
39	1	$-x^6 + 3x^5 + 5x^4 - 11x^3 - 10x^2 + 17x + 17$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of $(B_p^{(g)})_g$.

p	k	value	classes (a, b)
2	5	(1, -1)	(0, 11), (8, 19), (16, 27), (24, 3), (29, 0), (29, 8), (29, 16), (29, 24)
2	5	(-1, 1)	(13, 2), (13, 10), (13, 18), (13, 26)
2	6	(1, -1)	(4, 63), (20, 15), (36, 31), (52, 47)
2	6	(-1, -1)	(6, 61), (10, 49), (14, 5), (22, 13), (30, 21), (38, 29), (46, 37), (54, 45), (62, 53)
2	7	(1, -1)	(12, 7), (13, 4), (13, 36), (13, 68), (13, 100), (44, 39), (76, 71), (77, 20), (77, 52), (77, 84), (77, 116), (108, 103)

p	k	value	classes (a, b)
2	7	$(-1, 1)$	(109, 6), (109, 14), (109, 22), (109, 30), (109, 38), (109, 46), (109, 54), (109, 62), (109, 70), (109, 78), (109, 86), (109, 94), (109, 102), (109, 110), (109, 118), (109, 126)
2	7	$(-1, -1)$	(2, 41), (18, 57), (34, 73), (42, 81), (50, 89), (66, 105), (82, 121), (98, 9), (106, 17), (114, 25)
2	8	$(1, -1)$	(28, 23), (45, 124), (45, 252), (60, 55), (92, 87), (156, 151), (173, 28), (173, 92), (173, 156), (173, 220), (188, 183), (220, 215), (237, 12), (237, 76), (237, 140), (237, 204)
2	9	$(1, -1)$	(109, 44), (109, 172), (109, 300), (109, 428), (365, 108), (365, 236), (365, 364), (365, 492)
2	9	$(-1, -1)$	(58, 353), (122, 417), (186, 481), (218, 257), (250, 33), (314, 97), (378, 161), (410, 449), (442, 225), (506, 289)
2	10	$(1, -1)$	(124, 887), (380, 119), (508, 247), (636, 375), (813, 60), (813, 316), (813, 572), (813, 828), (892, 631), (1020, 759)
2	10	$(-1, -1)$	(26, 65), (90, 641), (154, 193), (282, 321), (346, 897), (474, 1), (538, 577), (602, 129), (666, 705), (794, 833), (858, 385), (986, 513)
2	11	$(1, -1)$	(301, 444), (301, 956), (301, 1468), (301, 1980), (1325, 188), (1325, 700), (1325, 1212), (1325, 1724)
2	12	$(1, -1)$	(252, 4087), (1276, 1015), (2300, 2039), (3324, 3063)
2	13	$(1, -1)$	(764, 4599), (1788, 5623), (2812, 6647), (3836, 7671), (4860, 503), (5884, 1527), (6908, 2551), (7932, 3575)
3	2	$(1, 1)$	(0, 4), (3, 1)
3	3	$(1, 1)$	(1, 19), (2, 14), (4, 19), (6, 7), (7, 19), (8, 20), (10, 19), (11, 23), (13, 19), (15, 25), (16, 19), (17, 2), (19, 19), (20, 5), (22, 19), (23, 8), (24, 16), (25, 19), (26, 11)
3	4	$(1, 1)$	(5, 17), (14, 26), (32, 44), (41, 53), (59, 71), (68, 80)
5	1	$(1, 1)$	(0, 3), (1, 2)
5	2	$(1, 1)$	(2, 9), (2, 10), (3, 23), (7, 10), (8, 13), (12, 9), (12, 10), (13, 3), (17, 9), (17, 10), (22, 9), (22, 10), (23, 8)
5	3	$(1, 1)$	(2, 56), (7, 34), (7, 106), (12, 81), (17, 106), (18, 18), (22, 56), (27, 56), (32, 34), (32, 106), (37, 81), (42, 106), (43, 93), (47, 56), (52, 56), (57, 34), (57, 106), (62, 81), (67, 106), (68, 43), (72, 56), (77, 56), (82, 34), (82, 106), (87, 81), (92, 106), (93, 118), (97, 56), (102, 56), (107, 106), (112, 81), (117, 106), (118, 68), (122, 56)
5	4	$(1, 1)$	(107, 34), (232, 34), (357, 34), (482, 34), (607, 34)
7	1	$(1, 1)$	(1, 4), (2, 0), (5, 3), (6, 5)
7	2	$(1, 1)$	(3, 9), (17, 9), (24, 9), (31, 9), (38, 9), (45, 9)
7	3	$(1, 1)$	(10, 254), (59, 254), (108, 254), (157, 254), (206, 254), (255, 254), (304, 254)
11	1	$(1, 1)$	(2, 6), (4, 0), (5, 6), (6, 7), (7, 7), (8, 2), (8, 8), (9, 5), (10, 6)

p	k	value	classes (a, b)
11	3	(1, 1)	(8, 1145), (19, 584), (30, 144), (41, 1156), (52, 958), (63, 881), (74, 925), (85, 1090), (96, 45), (107, 452), (118, 980), (129, 298), (140, 1068), (151, 628), (162, 309), (173, 111), (184, 34), (195, 78), (206, 243), (217, 529), (228, 936), (239, 133), (250, 782), (261, 221), (272, 1112), (283, 793), (294, 595), (305, 518), (316, 562), (327, 727), (338, 1013), (349, 89), (360, 617), (371, 1266), (382, 705), (393, 265), (404, 1277), (415, 1079), (426, 1002), (437, 1046), (448, 1211), (459, 166), (470, 573), (481, 1101), (492, 419), (503, 1189), (514, 749), (525, 430), (536, 232), (547, 155), (558, 199), (569, 364), (580, 650), (591, 1057), (602, 254), (613, 903), (624, 342), (635, 1233), (646, 914), (657, 716), (668, 639), (679, 683), (690, 848), (701, 1134), (712, 210), (723, 738), (734, 56), (745, 826), (756, 386), (767, 67), (778, 1200), (789, 1123), (800, 1167), (811, 1), (822, 287), (833, 694), (844, 1222), (855, 540), (866, 1310), (877, 870), (888, 551), (899, 353), (910, 276), (921, 320), (932, 485), (943, 771), (954, 1178), (965, 375), (976, 1024), (987, 463), (998, 23), (1009, 1035), (1020, 837), (1031, 760), (1042, 804), (1053, 969), (1064, 1255), (1075, 331), (1086, 859), (1097, 177), (1108, 947), (1119, 507), (1130, 188), (1141, 1321), (1152, 1244), (1163, 1288), (1174, 122), (1185, 408), (1196, 815), (1207, 12), (1218, 661), (1229, 100), (1240, 991), (1251, 672), (1262, 474), (1273, 397), (1284, 441), (1295, 606), (1306, 892), (1317, 1299), (1328, 496)